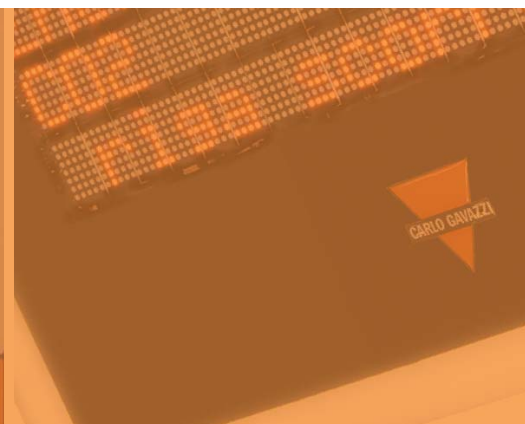




Solutions for Renewable Energy Systems

EcoEnergy

Carlo Gavazzi Automation Components



Providing components for automation

Carlo Gavazzi Automation is a multinational electronics group active in the design, manufacture and marketing of electronic equipment targeted at the global markets of industrial and building automation.

Our history is full of firsts and our products are installed in a huge number of applications all over the world. With 80 years of successful operation, our experience is unparalleled.

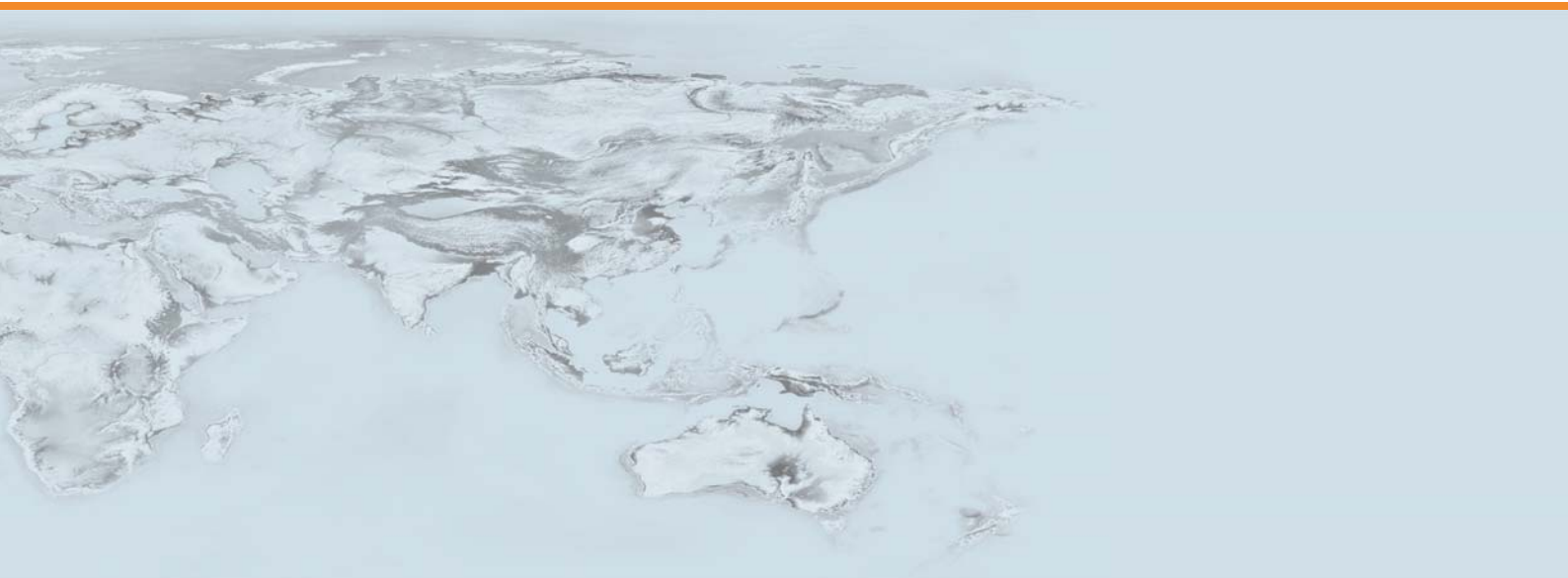
Our core competence in automation spans four product lines: Sense, Switch, Control and Fieldbus.

Our wide array of products includes sensors, monitoring relays, timers, solid-state relays, electronic motor controllers, energy management equipment and fieldbus systems.

We focus our expertise on offering state-of-the-art product solutions in selected market segments.

Our customers include original equipment manufacturers of packaging machines, plastic-injection moulding machines, food and beverage production machines, conveying and material handling equipment, door and entrance control systems, lifts and escalators, as well as heating, ventilation and air-conditioning devices.

We have our headquarters in Europe and numerous offices around the world. Our R&D competence centres and production sites are located in Denmark, Italy, Lithuania, Malta and the People's Republic of China. We operate worldwide through 21 wholly-owned sales companies and selected representatives in more than 65 countries, from the Pacific Rim in the East to the United States in the West.



Optimizing use of renewable energy sources

The increasing demand for energy, availability and costs of oil and issues of pollution make alternative sources of renewable energy essential. Sun and wind energy can be transformed into electricity that can be directly used or fed into the grid according to national regulations.

Renewable energy in combination with energy efficiency is the basis of sustainable development and respect for the environment in which we all live.

Our wide expertise is focused on developing and offering an EcoEnergy package of products and devices for the Renewable Energy market, specifically Photovoltaic and Wind power plants.

Our objective is to offer - to system integrators and customers - advanced and reliable equipment, expert technical support and efficient after-sales service.

PHOTOVOLTAIC SYSTEMS

The growing development of cost-effective solar energy technology has great potential to benefit our world. Solar technologies diversify the energy supply, reduce our dependence on fossil fuels, improve air quality and off-set greenhouse gas emissions.

Carlo Gavazzi offers complete and modular solutions to monitor and control the efficiency of Photovoltaic plants. Our product range for solar solutions includes inverters, solar battery chargers, energy meters and interface protections.

MICRO AEOLIC SYSTEMS

The term wind power describes the process by which the wind is used to generate mechanical power or electricity. Wind turbines convert the kinetic energy in the wind into mechanical power that can then be utilized. Carlo Gavazzi is able to offer a complete package for wind power generation.



Solar Energy

EcoEnergy
equipment



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The energy that the sun delivers to the earth's surface is equal to 89 PW (89 million GW). Several technologies have been developed in order to harness this energy: solar thermal, solar concentration plants and photovoltaic energy.

Solar thermal consists of installing particular panels in which water is heated by the sun in order to be used for heating systems or for the hot water itself to be used.

In *Solar concentration* plants the sunlight is captured by special mirrors that reflect the light onto a tank or pipes containing a special liquid which is heated up to several hundred degrees.

Photovoltaic is energy generated directly by PV panels. The electricity can then be used to charge batteries, by means of photovoltaic battery chargers, or is fed into the public utility grid through grid connected inverters.

The global electricity demand is roughly 18 TW, of which almost 90% is produced by using nuclear and fossil sources.

As the total PV capacity installed in 2009 was only 23 GW,

an insignificant figure if compared to the global electricity demand (0.0013%), there is enormous potential for the growth of this type of electricity generating system.

The CO₂ emissions of PV technology are negligible if compared to any fossil fuel technology energy production: this is also why PV is the fastest-growing renewable energy technology, and costs are expected to drop faster than those of other electricity sources.

The estimated lifetime of a PV module is 30 years. Furthermore, the module's performance is very high, providing over 80% of the initial power after 25 years, which makes photovoltaic a very reliable technology in the long term. Europe has set a target of 20% of energy to be produced by renewable sources by the year 2020. The PV share will most probably be the mainstream energy source in Europe and cover up to 12% of the electricity demand, with 390 GWp installed and an electricity production of around 460 TW/h (source: EPIA, 2010)



Solar Energy

Carlo Gavazzi intends to contribute to this change and for this reason has used its considerable experience in energy management for the development of a new range of products for renewable energy. The first product of this type, the DPC02, a grid monitoring relay, was launched in 2004. Today, apart from the PV modules, Carlo Gavazzi can provide most components for a PV plant installation, both for power generation and plant monitoring.

PV plants can be divided into 2 main types:

- Grid connected
- "Island" (Off grid)

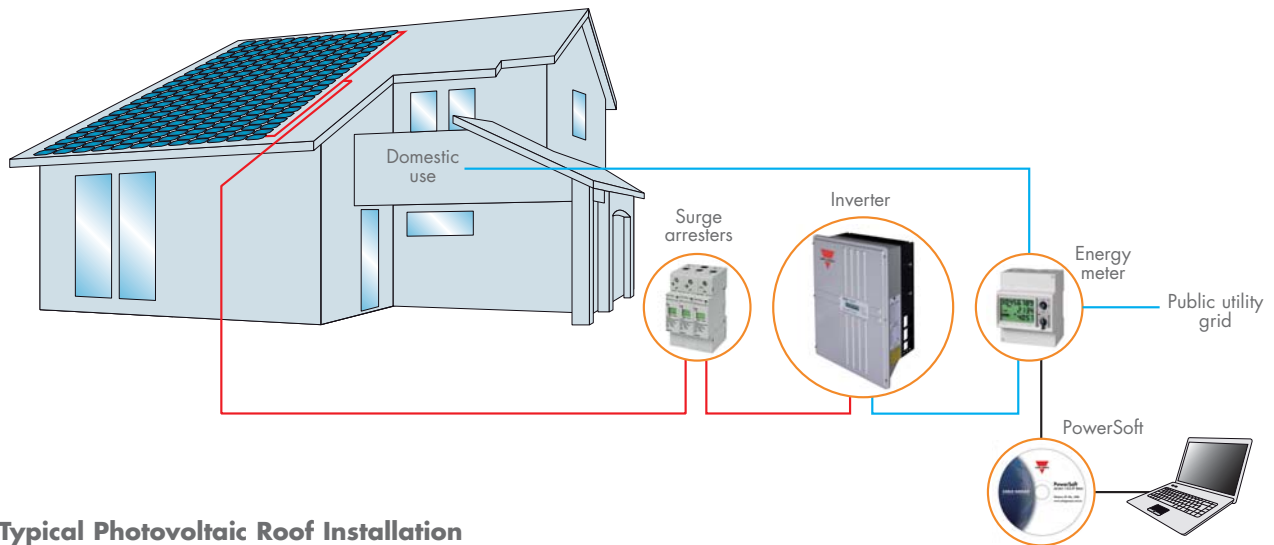
Grid Connected installations are the most common in commercial and residential areas. They are installed on roofs in residential, business or industrial sites. Modules are installed on the ground in photovoltaic farms. This is a typical installation for large portions of unused land in areas which are unpopulated but close to cities.

On both roof or ground, grid-connected PV installations the energy produced by the PV modules is delivered to the inverters and then fed into the public grid. In residential and business installations, if the demand is higher than the production, the

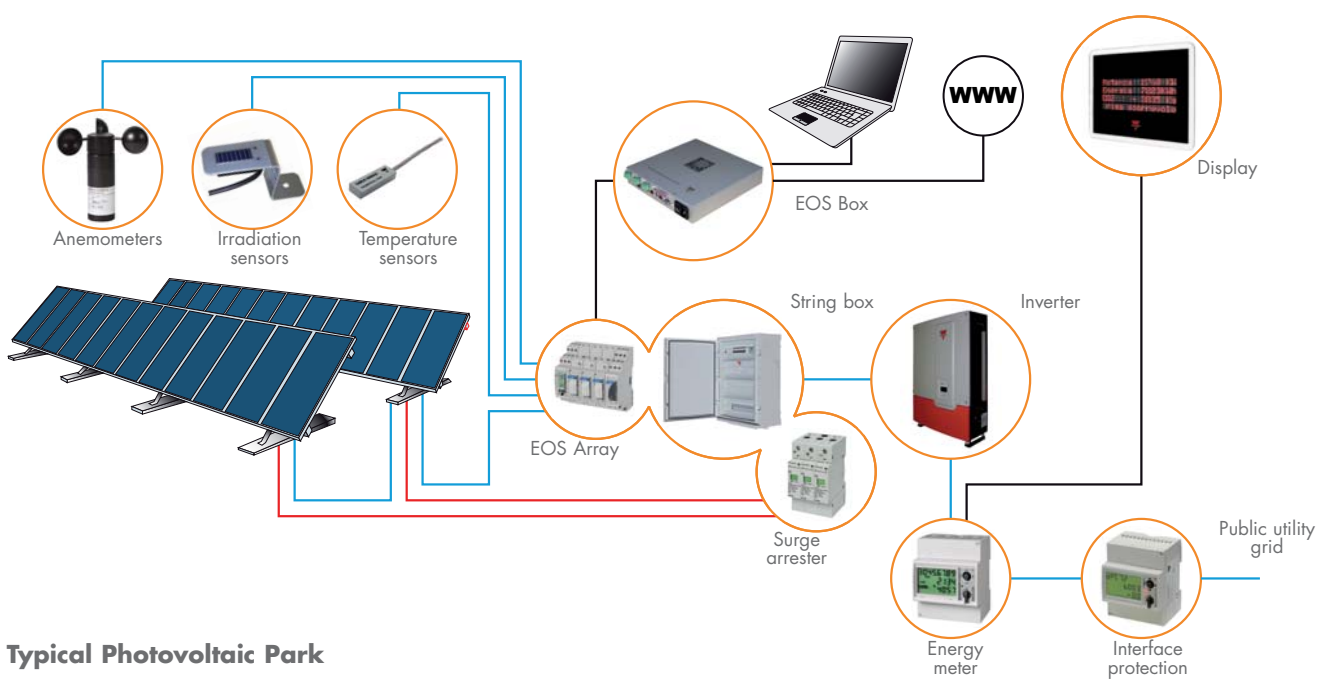
balance of missing energy is withdrawn from the grid. When the demand is lower than the produced energy, the excess is sold to the DNO (Distribution Network Operator).

In countries where there are "feed-in tariffs" there is a big advantage in selling the energy excess to the DNO, as the cost of the purchased energy is much less than what is paid for the energy sold to the utility. This mechanism enables the pay-back of investment in a shorter time. In the long run, no more support will be required to help the development of photovoltaic electricity. It is expected that photovoltaic will be competitive with electricity prices (at least in Europe, with a contribution of 60% of the global cumulative installed capacity) by 2020, with the so called "grid parity" (source: EPIA, 2010).

Island installations are typical of remote or rural areas far from any public utility grid. In these plants the energy produced by the PV modules is used to charge batteries. The energy of the batteries can then be used in order to power DC loads or it can be converted into AC voltage by means of battery powered inverters. Often the battery charging comes from more than one renewable source: photovoltaic, wind and/or water turbines. A combination of these sources increases the potential for battery charge in almost all conditions.



Typical Photovoltaic Roof Installation



Typical Photovoltaic Park

ISMG 1

Single-phase, grid tied PV inverter

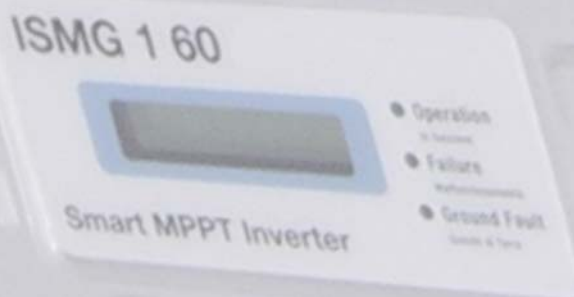
The ISMG 1 is a family of PV inverters dedicated to household and small installations. Having 2 or 3 inputs, each with an independent MPP tracker, it is especially suitable for typical East-West installations, or when part of the modules in some hours of the day might be shaded by chimneys, trees or other buildings, increasing the energy yield via the reduction or mismatching and partial shading losses.

The exclusive Smart MPPT technology ensures high yield, even in the case of mist or a cloudy sky.

MPP voltage from 100 VDC makes it particularly suitable for areas with low irradiation. The integrated data display with 3 indication LED's provides immediate monitoring of the inverter operational status, grid values and alarm messages. ISMG 1 inverters feature a superior cooling design with internal temperature control.

ISMG 1s are equipped with system monitoring and

interface protection, according to major European recommendations. The anti-islanding function guarantees secure disconnection in case of circuit isolation due to grid failure or disconnection, and avoids isolated operation. Installation is hassle-free, with IP65 protection, MC4 connectors and dual comm ports for easy chaining.



Maximize solar energy production

• Low Voltage MPP Input

MPP start voltage 100 VDC enables ISMG 1 inverters to start energy production when the irradiation is still very low.

• 2 or 3 independent String Inputs

The efficient MPP tracking can handle up to 2 or 3 independent MPP tracking inputs, each one controlled by exclusive smart MPP technology, which ensures the increase of energy up to 20%, thanks to the fully functional PV string control software.

• Transformerless with RCMU

These devices are transformerless but equipped with an RCMU circuit in order to monitor and limit the DC current injection on the grid.

• Superior cooling design

The ISMG 1 inverters have a very good thermal design. Their inner temperature is normally with full load around 55°C: other inverters normally work at 10°C. This highlighted working temperature makes the ISMG 1 more reliable and efficient in the long term, due to less electronic component stress. Derating starts at ambient temperature of 55°C. Few inverters can claim such thermal performance.

• Interface protection and Anti-Islanding

The integrated interface protection can be configured according to the major national DNO recommendations: VDE0126-1-1, DK5940, RD1660/RD661.

Timings and values of other recommendations can be configured in the firmware upon request.

An interface protection circuit monitors the grid and, in case of deviation from the preset values, the unit is disconnected from the grid by means of double redundant relays. The disconnection also occurs if the grid is not active, preventing the unit from working as an island (anti-islanding) and ensuring the safety of personnel involved in grid servicing.

• Built-in Comms

Built in RS232/RS485 serial communication (with 2 connectors for easy chaining) Modbus RTU protocol for remote monitoring.

• Easy commissioning

ISMG 1 inverters do not require set-up or programming. The bracket can be locked to the inverter for theft protection. IP65 makes it suitable for indoor as well as outdoor installation.

Photovoltaic DC Input Data

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Nominal DC power	3.15 kW	3.48 kW	4.0 kW	4.85 kW
Max. DC power	3.6 kW	4.0 kW	4.64 kW	5.38 kW
Max. DC voltage	500 V			
MPP voltage range	100 to 450 V			
Max. DC current per string	10 A			
Number of MPP trackers	1~2		1~3	
Overvoltage protection	Yes			
Ground fault monitoring	Yes			
Polarity safeguard	Short circuit diode			
Surge absorption	MOV varistor			

AC Output Data

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Rated output AC power	2.99 kW	3.3 kW	3.8 kW	4.6 kW
Max. AC power	3.0 kW	3.8 kW	4.4 kW	5.1 kW*
Power factor	> 0.99			
Distortion factor (THD)	< 3%			
Grid connection	Single phase; True sine-wave			
AC voltage range	rated 230 VAC			
AC nominal current	13 A	14.34 A	16.52 A	20 A
Frequency range	rated 50 Hz			
Safety class	I			
All pole sensitive RCMU	Yes			

* German version limited to 4.99 kW

General Data

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Max. efficiency	96.2% @ 350 VDC		96.3% @ 350 VDC	
EU efficiency	95.4% @ 350 VDC		95.1% @ 350 VDC	95.4% @ 350 VDC
Stand-by consumption	< 10 W			
Night consumption	0 W			
Protection device	Grid monitoring system			
Anti-islanding monitoring	Yes			
Grid monitoring	Integrated interface protection, Programmable parameters according to local DNO recommendations			

Environmental Data

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Operating temperature @ 350 VDC	-10°C to +60°C / 14°F to 140°F without derating; Shut-down at 75°C/167°F		-10°C to +55°C / 14°F to 131°F without derating; Shut-down at 75°C/167°F	
Storage temperature	-25°C to +80°C / -13°F to 176°F			
Humidity	0 to 95% (without condensation)			
Cooling	Forced ventilation (2 x IP54 external fans)			
Protection degree	IP65 (according to DIN EN60529)			
Noise level	< 40 dB			

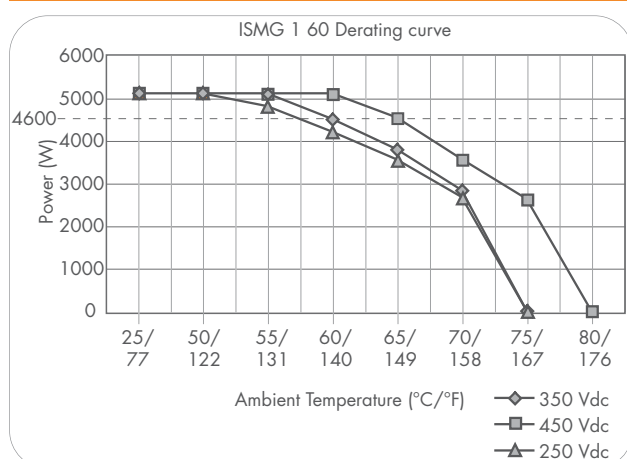
Mechanical Data

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
User interface	2 x 16 LCD Data display and 3 monitoring LEDs			
AC connectors	1 x Wieland			
DC connectors	2 x 2 Multicontact (MC4 series)			3 x 2 Multicontact (MC4 series)
Serial interface connectors	2 x RJ45			
Housing material	Powder coated aluminium			
Weight	22.5 kg/49.60 lb			23.0 kg/50.70 lb

Standard Norms and Certifications

Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Safety Standard	EN50178			
EMC capability	EN61000-3-2, EN61000-3-3 EN61000-3-11, EN61000-3-12 EN61000-6-2, EN61000-6-3			
Grid monitoring	ES: according to RD 1663/2000; RD 661/2007 IT: according to DK5940 Ed. 2.2 April 2007 FR, EN, DE: according to VDE0126-1-1			

Temperature Range



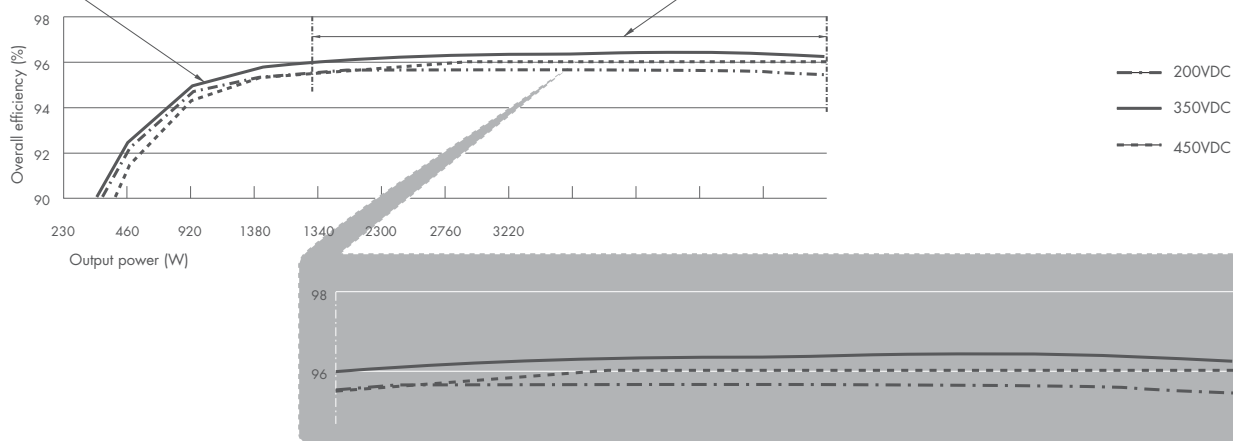
Cooling fan control

Command	Heat-sink temperature (°C/°F)
FAN start	50/122
FAN stop	45/113
Derating temperature	72/161.6
Inverter shut-down temperature	80/176

Efficiency

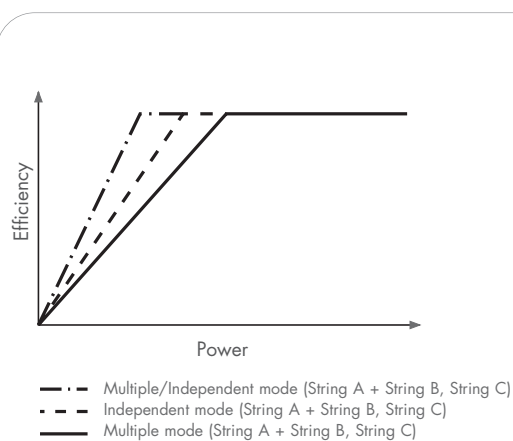
At 1 kW output power, ISMG 1 inverter has an efficiency of 95% @ 200 VDC and 350 VDC

ISMG 1 inverter has a consistent efficiency greater than 96% from 1.7 kW to 5.1 kW (350 VDC)

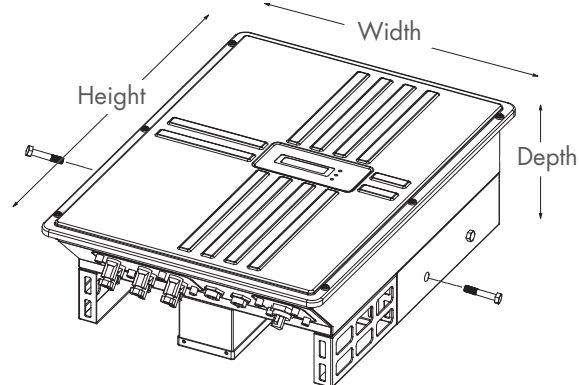


Model	ISMG 1 45 PL3	ISMG 1 45	ISMG 1 50	ISMG 1 60
Efficiency 5% P _{nom}	84.42%	84.42%	85.70%	87.65%
Efficiency 10% P _{nom}	91.19%	91.19%	90.94%	92.04%
Efficiency 20% P _{nom}	94.27%	94.27%	94.43%	94.86%
Efficiency 30% P _{nom}	95.37%	95.37%	95.36%	95.62%
Efficiency 50% P _{nom}	96.04%	96.04%	95.58%	96.11%
Efficiency 100% P _{nom}	96.28%	96.28%	96.07%	96.10%

Multiple/Single Panel String Mode



Dimensions



Model	All Models
H x W x D mm/inches	580 x 422 x 182 (+6.5 mounting support) 22.83" x 16.61" x 7.17" (+0.26" mounting support)

ISMGT 1

HF transformer, grid tied PV inverter

The ISMGT 1 is a family of PV inverters dedicated to household and small installations. They are, equipped with a high frequency transformer that galvanically isolates the DC input section from the AC part, making them suitable for all those applications where it is mandatory to isolate input from output.

They can be used with thin film modules that are usually connected with positive or negative grounding.

A single MPP tracker - up to 4 strings inputs - can be jumpered inside the special connection box.

The connection box, also equipped with a front power disconnect, provides the grounding inside the jumpers.

ISMGT 1 inverters feature efficiency up to 97%, one of the highest performances among the transformer inverters of their class.

Integrated data display with 3 indication LEDs provides immediate monitoring of the inverter operational status, grid

values and alarm messages. ISMGT 1 inverters feature superior cooling design with internal temperature control.

These inverters, as well as other Carlo Gavazzi grid tied inverters, are equipped with system monitoring and interface protection, according to major European recommendations. The anti-islanding function guarantees secure disconnection in case of circuit isolation due to grid failure or disconnection, and avoids isolated operation.



Optimize performance of PV plants

• Wide MPP input range

The MPP of this series of inverters has a range of 350 V, allowing operation from 200 up to 550 VDC.

• Power disconnect and fuse

The front disconnect enhances the safety of this device. It allows the cutting of both AC and DC connections and enables the opening the front cover only when power is OFF. On the bottom of the connection box there is also a fuse holder for the ground fuse.

• 3 or 4 string inputs

The inverter is equipped, according to the model, with 3 or 4 string inputs which are managed by a single MPPT circuit. The inputs can also be configured by jumpers to be positive or negative grounded inside the connection box.

• Superior cooling design

The ISMGT inverters have a very good thermal design. The inner temperature is electronically controlled and the fans are activated when necessary. Derating ambient temperature starts at 58°C, which makes it the best choice for hot places.

• Built-in comms

Built in RS232/RS485 serial communication (with 2 connectors for easy chaining) Modbus RTU protocol for remote monitoring.

• High frequency transformer

The DC input side is galvanically insulated from the AC output side by mean of high frequency transformer. It is possible to use all types of modules with the inverter: poly, mono crystalline or amorphous.

• Interface protection and Anti-islanding

The integrated interface protection can be configured according to the major national DNO recommendations: VDE0126-1-1, DK5940, RD1660/RD661 or even G.83 Timings and values of other recommendations can be configured in the firmware upon request.

An interface protection circuit monitors the grid and, in case of deviation from the preset values, the unit is disconnected from the grid by means of double redundant relays. The disconnection also occurs if the grid is not active, preventing the unit from working as an island (anti-islanding) and ensuring the safety of personnel involved in grid servicing.

Photovoltaic DC Input Data

Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50
Nominal DC power	3.0 kW	4.0 kW	4.2 kW	5.2 kW
Max. DC power	3.1 kW	4.1 kW	4.3 kW	5.3 kW
Max. DC voltage	600 VDC			
MPP voltage range	200-550 VDC			
Full MPP range	350 VDC			
Max. DC current per string	15 A			
Number of MPP trackers	1			
Max. No. of parallel strings for MPP	3		4	
Overvoltage protection	Yes			
Ground fault monitoring	Yes, with led indication and integrated fuse			
Polarity safeguard	Diode + transformer insulation			
Surge absorption	Replaceable MOV varistors			

AC Output Data

Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50
Max. AC power	2.8 kW	3.8 kW	4.0 kW	5.0 kW
Power factor	> 0.99 @ nominal power			
Distortion factor (THD)	< 3%			
Grid connection	Single phase, 230 VAC			
AC voltage range	Depending upon setting 230 VAC nominal			
Max. AC current	12.2A	16.6 A	17.4 A	21.8 A
Frequency range	Depending upon setting 50 Hz Nominal			
Safety class	I			
All pole sensitive RCMU	Yes			

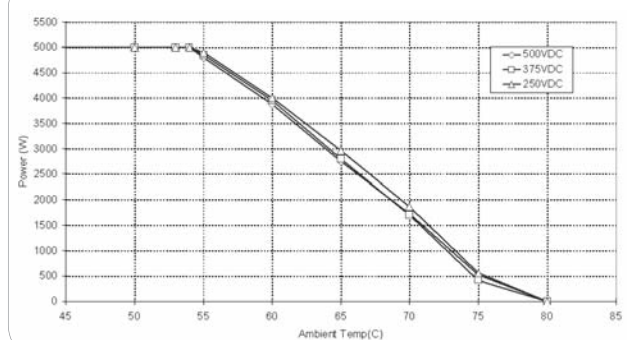
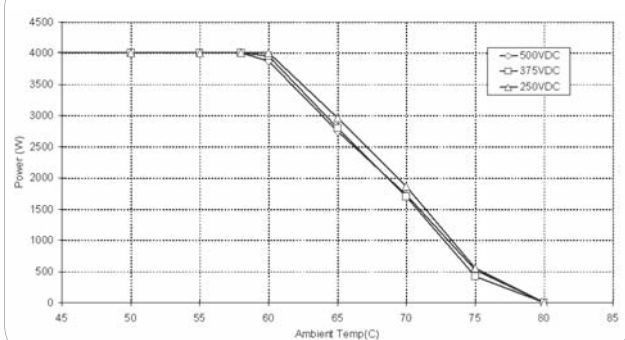
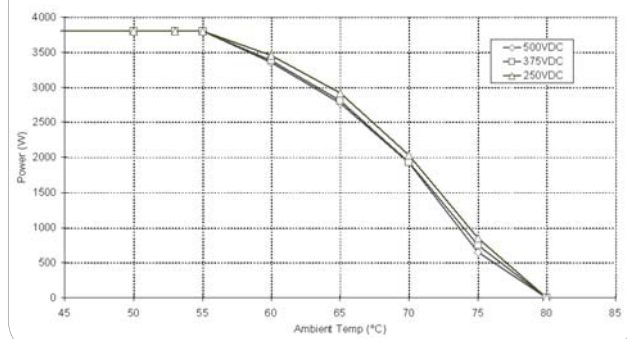
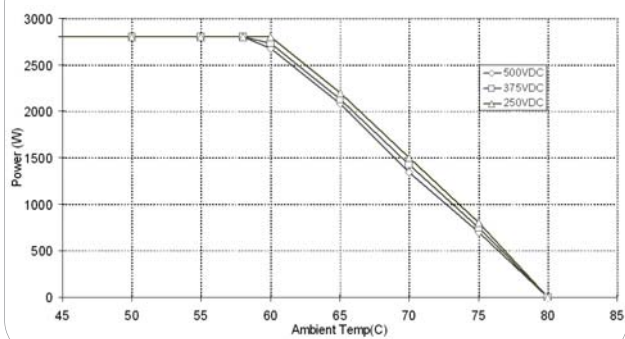
General Data

Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50
Max. efficiency	96.4%		96.5%	
EU efficiency	95.4%	95.7%	95.8%	
Night consumption	0.5 W			
Protection device	Grid monitoring system			
Anti-islanding monitoring	Yes			
Grid monitoring	Integrated interface protection with programmable parameters according to local DNO recommendations			

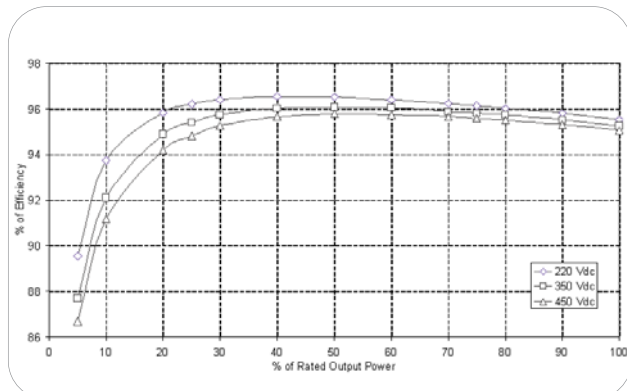
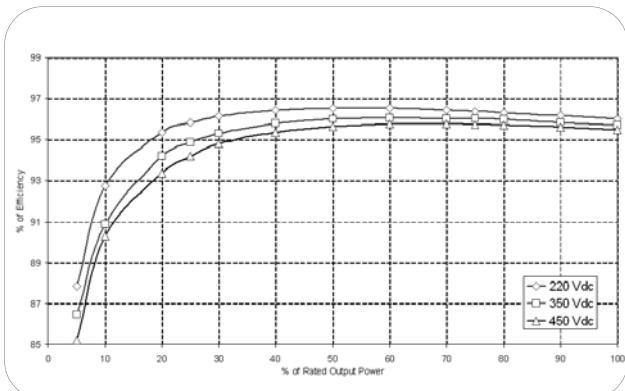
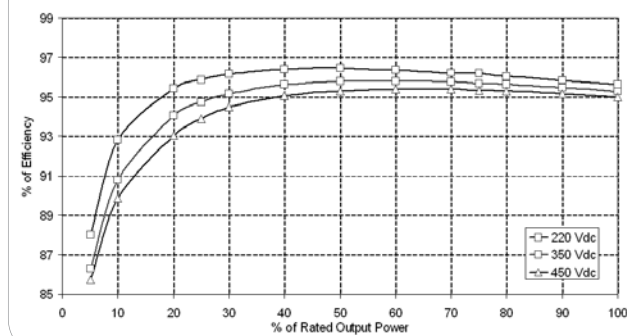
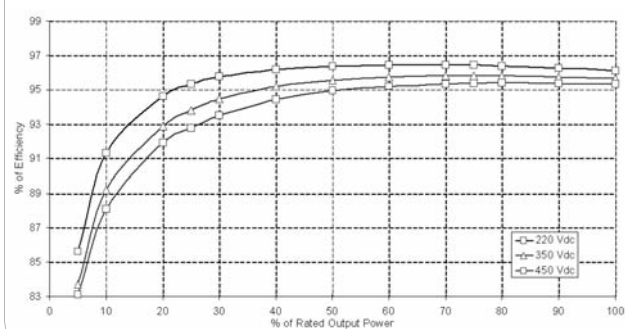
Environmental Data

Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50		
Operating temperature range	-25°C to +65°C / -13°F to 149°F					
Max. full power operating temperature without derating	+58°C / 136°F	+55°C / 131°F	+58°C / 136°F	+54°C / 129°F		
Cooling	Convection		Controlled forced ventilation			
Protection degree	IP44					
Installation location	Indoor / Outdoor protected					

Derating Curves



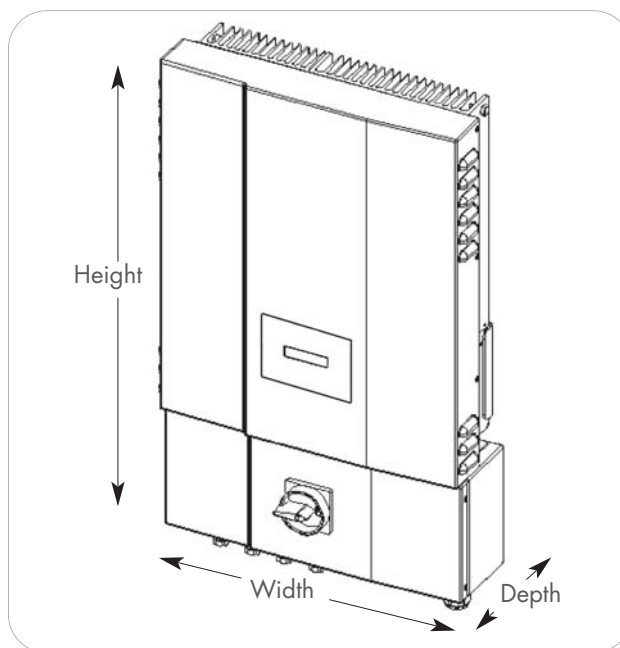
Efficiency Curves



Mechanical Data

Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50
Inverter weight	23 kg		28 kg	
Shipping weight	27 kg		32 kg	

Dimensions



Model	ISMGT 1 28	ISMGT 1 38	ISMGT 1 40	ISMGT 1 50
H x W x D mm/inches	768x454x175 (+6.5 mounting supp.) 30.2"x17.9"x6.9" (+0.26" mounting supp.)		768x454x210 (+6.5 mounting supp.) 30.2"x17.9"x8.3" (+0.26" mounting supp.)	

ISMG 3

3 phase, grid tied PV inverter 17 to 22 kW

The Carlo Gavazzi ISMG 3 series PV inverters convert the energy generated by the solar modules into true sinewave balanced three phase 400 VAC power to be fed into the utility grid.

The anti-islanding and the grid monitoring protections ensure system safety and compliance with the most relevant national recommendations for grid connected generating systems.

2 string inputs, managed by separate MPPT circuits and the very wide MPP range, allow system design flexibility and ensure power generation in many different irradiation situations.

All units are equipped with built in serial communication for the remote monitoring of the production data for daily and monthly produced energy. It is possible to use standard remote monitoring units in order to read and show the data stored in the device.

The memory can store up to 12 months of data and the integrated graphic display can show the actual measurements, or last 100 alarms by text message or

stored produced energy data. The display backlight changes colour (red, green, white) according to the operational status, showing at a glance, even from a far, the inverter status.

To increase safety and make installation easier, all ISMG 3 inverters are also equipped with a DC switch, mounted on the side, with which makes it possible to disconnect the input strings at any time.

They can be installed either internally or externally, on walls or on the ground, with their special "floor mounting stand" that can be ordered separately.



Feed power effectively into the grid

• Wide high voltage MPP input

The MPP range is very wide, from 300 VDC to 850 VDC, maintaining conversion efficiencies above 97% throughout all input voltage ranges.

• 2 Input strings

The ISMG 3 is equipped with 2 efficient trackers, with "Smart MPP Technology". The input strings can be connected separately, enabling flexibility in module selection and PV system sizing, or connected in parallel with one MPP tracker to achieve higher efficiency.

• Easy commissioning

ISMG 3 inverters do not require set-up or programming. For wall mounting a special bracket is provided. For floor mounting a kit can be provided upon request. The bracket can be locked to the Inverter for theft protection.

• Graphic display and Minilogger

128 x 64 Graphic display with 5 keys integrated keypad, enables the reading of actual production data: Input string voltages (VDC), string currents (A), output power (kW), grid voltage (VAC), output current (A), produced energy (kWh). Connected to the integrated minilogger are the trend graphs of energy produced daily (kWh), during the last 31 days, or the energy produced monthly (kWh) during the last 12 months.

• Interface protection and Anti-islanding

The integrated interface protection can be configured according to the major National DNO recommendations: VDE0126-1-1, DK5940, RD1660/RD661. G.83 (with external transformer). Timings and values of other Recommendations can be configured in the firmware upon request.

The interface protection circuit monitors the grid and in case of deviation from the preset values the unit is disconnected from the grid by means of double redundant relays. The disconnection also occurs if the grid is not active, preventing the unit from working as an island (anti-islanding) and ensuring the safety of personnel involved in grid servicing.

• Built-in comms

Built in RS232/RS485 serial communication (with 2 connectors for easy chaining) Modbus RTU protocol for remote monitoring.

• High temperature operation

17 kW inverters work up to 50°C and 22 kW up to 45°C, without derating. Cooling fans are electronically controlled.

• Transformerless with RCMU

These devices are transformerless but equipped with an RCMU circuit in order to monitor and limit the DC current injection on the grid.

Photovoltaic DC Input Data

Model	ISMG 3 15	ISMG 3 20
Nominal DC power	17.9 kW	23.1 kW (21 kW*)
Max. DC power	17.9 kW	23.1 kW (21 kW*)
Max. DC voltage	850 V	
MPP voltage range	300 to 850 V	
Max. DC current per string	2 x 21.6 A (43.2 A)	2 x 28.9 A (57.8 A)
Number of MPP trackers	2	
Overvoltage protection	Yes	

AC Output Data

Model	ISMG 3 15	ISMG 3 20
Max. AC power	17 kW	22 kW (19.9 kW*)
Power factor	> 0.99% @ Rated Power	
Distortion factor (THD)	< 5%	
Grid connection	True sine-wave Balanced, 3 Phase	
AC voltage range	3 x 400 VAC (320 to 460 VAC) with progr. interface settings according to local DNO recommendation	
Max. AC current	24.7 A	31.9 A (31.88 A)*
Frequency range	50 Hz to 60 Hz with progr. interface settings according to local DNO recommendation	

* This limit is specific for the IT version (according to DK5940 norm)

General Data

Model	ISMG 3 15	ISMG 3 20
Max. efficiency	97.4%	97.6%
EU efficiency	97.0%	97.2%
Night consumption	1 W	
Protection device	Grid monitoring system (values according to local DNO recommendation)	

Environmental Data

Model	ISMG 3 15	ISMG 3 20
Operating temperature with derating > 50°C	-25°C to +60°C / -13°F to 140°F	
Max. acceptable temperature @ P _{nom}	+50°C / 126°F	+45°C / 121°F
Storage temperature	-25°C to +70°C / -13°F to 158°F	
Cooling	Integrated fans with operation controlled by software	
Protection degree	IP55 (according to DIN EN60529)	
Noise level	< 50 dB	< 55 dB

Standard Norms and Certifications

Model	ISMG 3 15	ISMG 3 20
Safety Standard	EN50178	
EMC capability	EN61000-3-2, EN61000-3-3, EN61000-3-11, EN61000-3-12 EN61000-6-2, EN61000-6-3	
Grid monitoring settings	EN VDE0126-1-1, ES RD 1663/2000, IT DK5940 Ed. 2.2 April 2007 DE VDE0126-1-1, FR VDE0126-1-1	

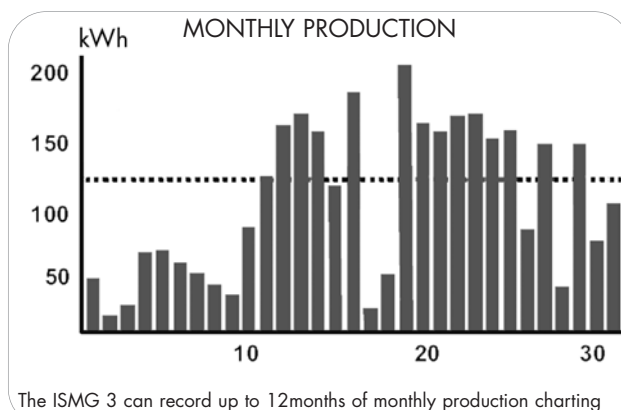
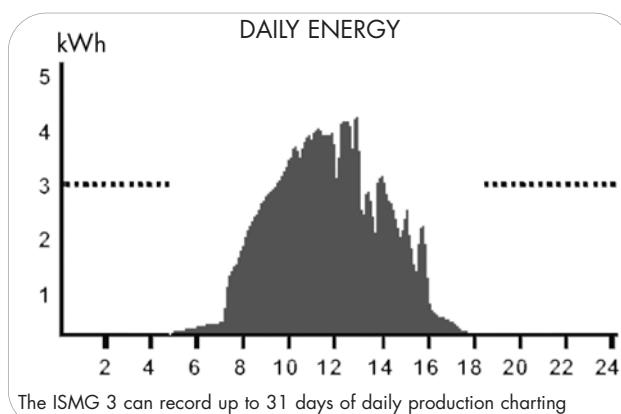
User Interface

Model	ISMG 3 15	ISMG 3 22
Display	128 x 64 LCD graphic display with variable backlight colour	
Keypad	4 key membrane: UP; DOWN; ENTER; ESC	
AC connectors	1 x Wieland	
DC connectors	2 x 2 Multicontact (MC4 series)	
Serial interface connectors	2 x RJ45 (for chain connection)	

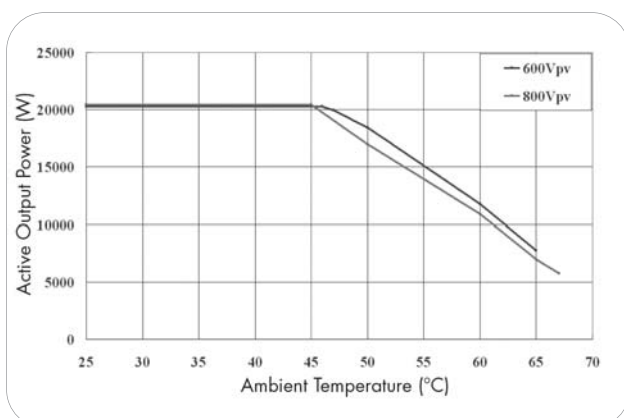
Accessible Display Data

Parameter	Unit
Output Voltage for each phase	VAC
Output Freq.	Hz
AC power fed into the grid	W
Produced AC Energy Today	kW/h
Produced AC Energy Total	kW/h
CO2 saved	kg
PV V string A	VDC
PV I string A	A
PV Power A	W
PV V string B	VDC
PV I string B	A
PV Power B	W
Actual VAC grid L1	VAC
Actual Iac grid L1	A
Actual Pac Grid L1	W
Actual VAC grid L2	VAC
Actual Iac grid L2	A
Actual Pac Grid L2	W
Actual VAC grid L3	VAC
Actual Iac grid L3	A
Actual Pac Grid L3	W

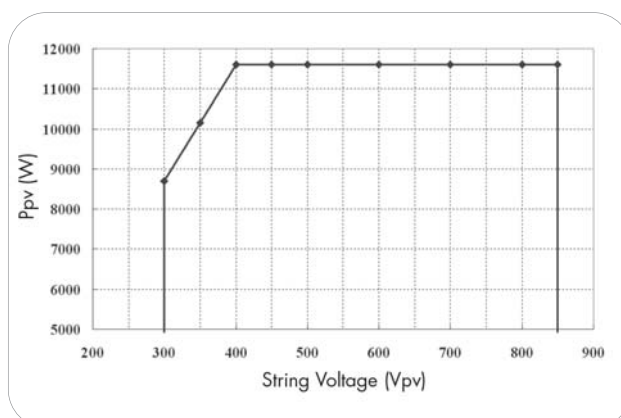
Display Graphic Functions



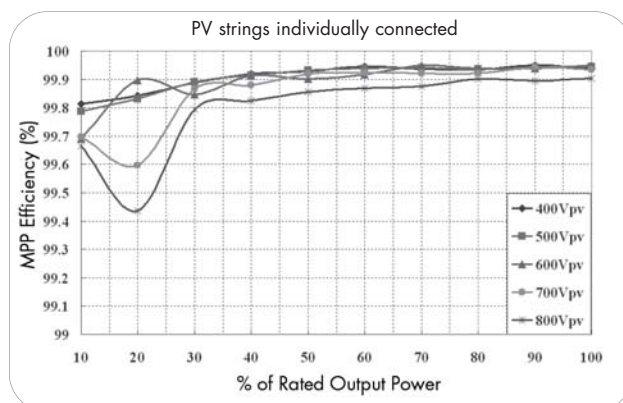
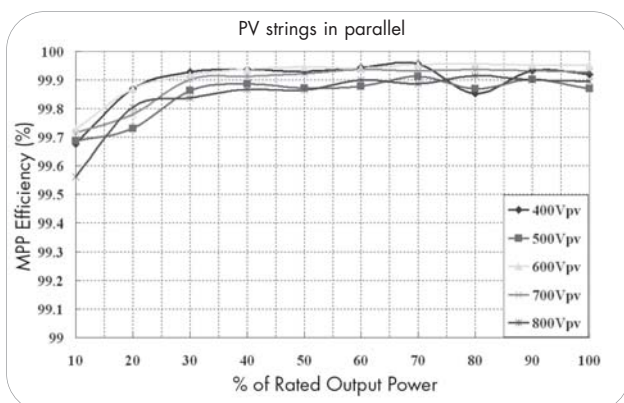
Temperature Behaviour



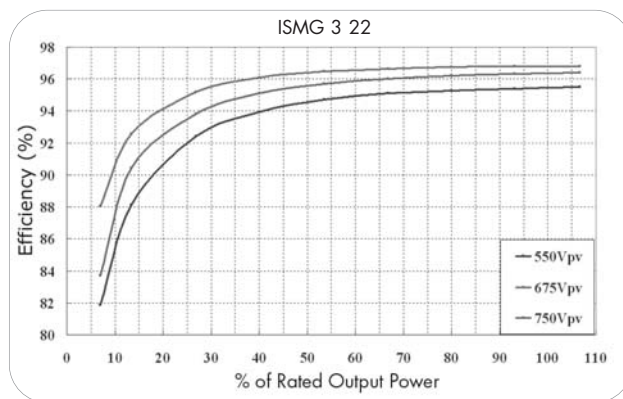
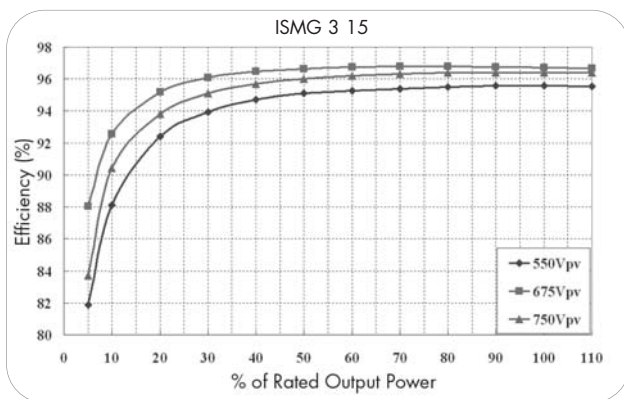
DC Curve



MPP Efficiency Diagrams



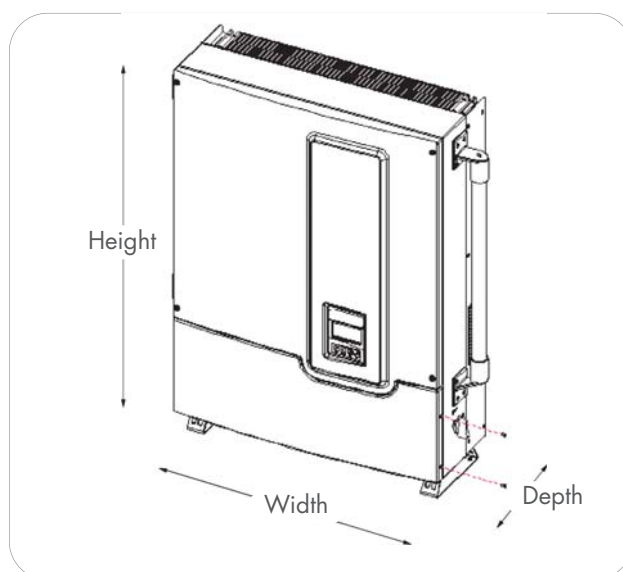
Efficiency Diagrams



Mechanical Data

Model	ISMG 3 15	ISMG 3 22
Housing material	Powder coated aluminium	
Weight	74.5 kg/164.2 lb (85 kg/187.4 lb shipping weight)	

Dimensions



Model	All Models
H x W x D mm/inches	890.5 x 751 x 256 (+6.5 mounting support) 35" x 29.57" x 10" (+0.26" mounting support)

Sun-View

Data logger web server

The Sun-View is an embedded PC which has been specifically programmed, with a linux based operating system, in order to operate as a data logger for Carlo Gavazzi solar inverters and as an Internet web server. It can be connected to a maximum of 6 inverters

Carlo Gavazzi has a consolidated experience in the developing and manufacturing monitoring solutions for medium and large photovoltaic installations from 100 kWp upwards. The Sun-View has also been designed with the goal of providing monitoring solution for the residential, small business and commercial markets.

The Sun View can be used for installations up to 120 kWp with Carlo Gavazzi inverters: but the installations with one single-phase inverter or with more than one tri-phase inverter with total power up to 30 / 50 kWp are the best match.

It is configurable either locally or remotely via the Internet. The current or memorized data, once the device has been

programmed and connected to the Web, can be visualized by means of the most common browsers.

The data is stored into a Flash Card but can also be stored as a backup on the PC.

Data can also be displayed on a PDA or Smartphone in a format suitable for their screens.



Keep your performance under control

• Communication

Being a Carlo Gavazzi product, the Sun-View is designed and programmed for operation and communication with all ISMG and ISMGT series inverters. It is also possible to connect a Carlo Gavazzi Energy Meter EM24 or EM26 for the recording of the energy actually fed into the grid by the whole installation.

• Displayed data

Real Time. For each of the active inverters the different values read in real time are displayed in an analogic dashboard like format. The current inverter status is also displayed. The trend of the current day yield is displayed in a graphic format: when the mouse cursor is passed along the graph the detailed values for each instant appear.

- Daily and monthly trend graphs. By selecting the day of the month, the graphs of the daily trend of each of the installed inverters are shown. The production trend graphs of the month, in which the selected day is, are also displayed. Furthermore, in these graphs a line indicating the calculated average monthly production is displayed.

- The data can be exported in CSV format. When exported, the total yield data, also individually for each inverter is available in a table format.

From the graphic pages it is possible to switch to the graphs of instant power, voltage and current indication for each of the strings.

• Economic performance

The Sun-View provides the daily and total economic yield in €.

• Annual report display

By selecting a specific year, the annual report is shown with the indication of yield. From this page it is possible to switch from one month to the other of the selected year.

• Parameters list

The operating parameters of the inverters, the same data that appear on the inverter display, are periodically sent to the Sun-View and thus displayed in real time in a table format on a specific page. By pressing a switch on the Sun-View interface on the screen the organisation of the page is optimized in a form suitable for the display on PDA screens.

• Data backup on PC

The Sun-View features FTP and SMB servers through which it is possible to access the device remotely in order to backup the data memorised in the internal flash memory.

General Data

Model	SVISMG60001
Type	Embedded PC
Operating system	Linux Platform
Operation	
Processor	AMD LX800 500 MHz
Chipset	AMD Geode LX800/ CS5536
BIOS	Flash memory 512 KByte (Award-BIOS)a
Memory	
RAM	256 MByte SDRAM
CompactFlash™ memory	1 GB (>20 years)
RS485	1 USB/RS485 converter supplied
Internet connection	1 Ethernet 10/100 MBit/s (via VT6105M) RJ-45
USB	2 (The USB ports can be used, by means of the proper converter for RS485 communication to the inverters or to an EM24 or EM26 meter)
Date / Time	RTC with back-up battery

Mechanical Data

Mounting	Wall or table
Weight	500 gr
Dimensions (HxWxD)	30 x 178 x 174 mm
Enclosure Material	Zinc plated steel

Environmental Data

Operating temperature	0 to +40°C
Humidity	0 to 95% RH
Protection degree	IP40

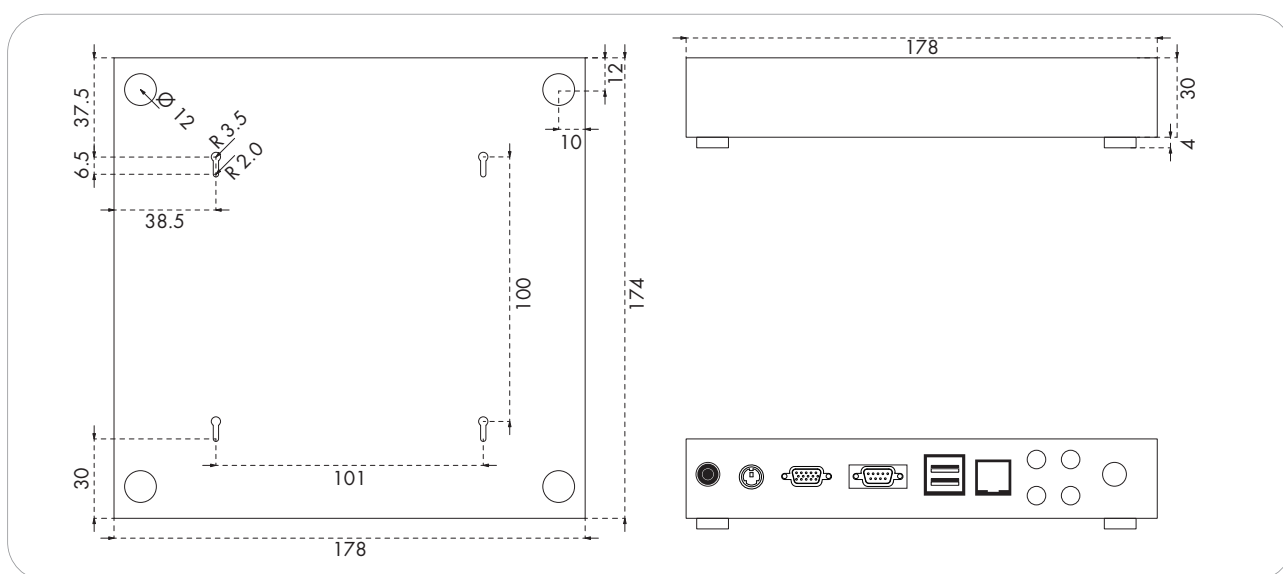
Power Supply Data

Supply voltage	12 V (power supply included)
Power Consumption	5 W

Standard Norms and Certifications

CE	Acc.to 89/336/CE and 73/23/CE complies with EN 61000-6-3 EN 61000-6-1 EN 60950-1
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Dimensions (mm)



Eos

Eos Array, Eos Array Lite, Eos Box

The Carlo Gavazzi solution for solar control develops with Eos Array and Eos Box, depending upon the requirements and complexity of the photovoltaic plant. Eos Array and Eos Array Lite are systems comprised of individual modular elements that, when interacting with one another, provide efficient local control of the solar plant. Whether it's a small/medium power or a high power plant, you are assured there is effective information management. The Eos Array provides a compact, simple, versatile and innovative solution, it's modular and can be composed of: VMU-M, the master unit and data logger; VMU-S, the string controller; VMU-P, the environment variable unit and VMU-O, the I/O unit: a mixed combination of up to 16 units in total. Eos Array Lite is the answer to those photovoltaic applications where a less sophisticated monitoring solution is needed. It is based on the same overall concept as Eos Array in terms of modularity and integrated string fuse protection, but its

focus is to measure and control only the string current and voltage. Eos Box is a web server that controls and supervises the whole installation, acquiring information from Eos Array groups, inverters, energy meters and interface protection. The Eos Box provides information in a quick and automatic fashion via the internet, so that the data is available wherever you are.



Control and supervise the whole system

• Eos Array:

- Measurement and control functions split into independent modules in order to improve the critical system reliability.
- String efficiency and BOS efficiency measurement are fundamental parameters to evaluate the yield of the photovoltaic plant.
- Local alarm management for fast and easy identification.
- Integrated fuse protection for cost and installation time saving.
- Easy application adaptability - the customer can choose according to his project philosophy which unit/s best suits his application giving him the freedom to find different solutions to his application problems.
- Features expansion capability based on usage of different modules and quantities so as to meet application needs from small (10 kWp) to large solar PV plants.
- Multicolour LEDs provide immediate feedback on variables, alarms, fuse status and communication, showing the relevant details on the display.

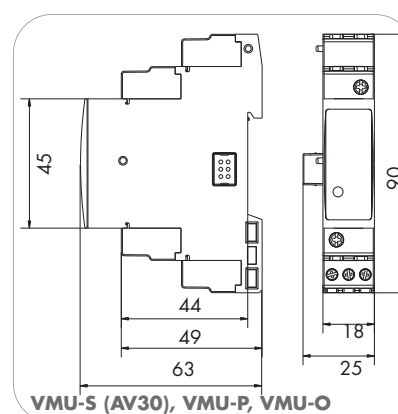
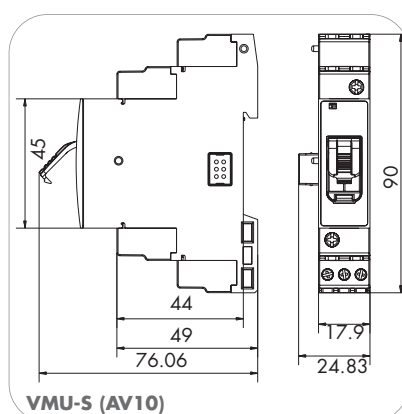
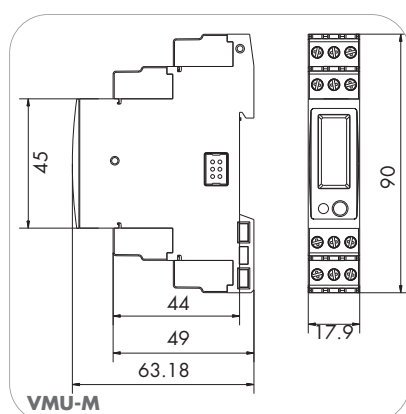
• Eos Box:

- Cost effective yield efficiency and plant failure warning with additional manual or automatic database download for easy plant history analysis.
- Plant failure warning by means of e-mail and SMS in the case of external connection via modem. Plant database download by means of EosArraySoft.
- Notification of abnormal conditions as soon as they occur for a prompt service intervention.
- Comprehensive control solution capable of showing efficiency, yield graphs and the management of any useful information which is accessible by the user from any place in the world just by using his PC browser.

Eos Array - Functions and Features

		Eos Array Lite	Eos Array
VMU-M Local Bus Management unit	All		
	Modular concept	Yes	Yes
	12 to 28 VDC power supply	Yes	Yes
	RS485 communication port (max. 115.2 kbit/s)	Yes	Yes
	Programming of communication speed and address	Yes	Yes
	Local display with programming pushbutton	Yes	Yes
	Measurement of single string current from VMU-S only	Yes	Yes
	Alarm management on measured variables	Yes	Yes
	Local bus-configuration errors control	Yes	Yes
	Current string monitoring	Yes	
	Power string monitoring		Yes
	String efficiency		Yes
	Event-logger: variables, functions and system alarms		Yes
	Data-logger: V, A, W, Wh, PV cell temperature, air temperature, irradiation		Yes
	First digital input or temperature input: Pt100/Pt1000 (3-wire)		Yes
	Second digital input or temperature input: Pt100/Pt1000 (3-wire)		Yes
VMU-S String Control unit	Local status monitoring by means of LED		Yes
	6-DGT data format for energy		Yes
	4-DGT data format for instantaneous variables	Yes	Yes
	Integrated fuse holder	Yes	Yes
	String voltage measurement up to 1000 V (0.5% RDG)	Yes	Yes
VMU-S String Control unit	String current measurement up to 16 A (0.5% RDG)	Yes	Yes
	String current measurement up to 30 A (0.5% RDG)		Yes
	String power measurement (1.0% RDG)		Yes
	String energy measurement (Class 1)		Yes
	Fuse blow alarm		Yes
	Fuse temperature alarm	Yes	Yes
	Wrong connection (reverse current or voltage)	Yes	Yes
	Configuration errors monitoring	Yes	Yes
	Irradiation sensor input: max. 120 mV	Yes	Yes
	First temperature probe input: Pt100/Pt1000 (3 wire)	Yes	Yes
VMU-P Environment Measuring unit	Short/open circuit on probe inputs	Yes	Yes
	Second temperature probe input: Pt100/Pt1000 (3 wire)		Yes
	Wind speed sensor measuring input		Yes
	Configuration errors monitoring	Yes	Yes
	First relay output activated by local alarm	Yes	Yes
VMU-O Inputs/Outputs unit	First relay output managed as remote command		Yes
	Second relay output activated by local alarm		Yes
	Second relay output managed as remote command		Yes
	First digital input		Yes
	Second digital input		Yes
	Configuration errors monitoring	Yes	Yes

Eos Array - Dimensions (mm)



Eos Box - General Data

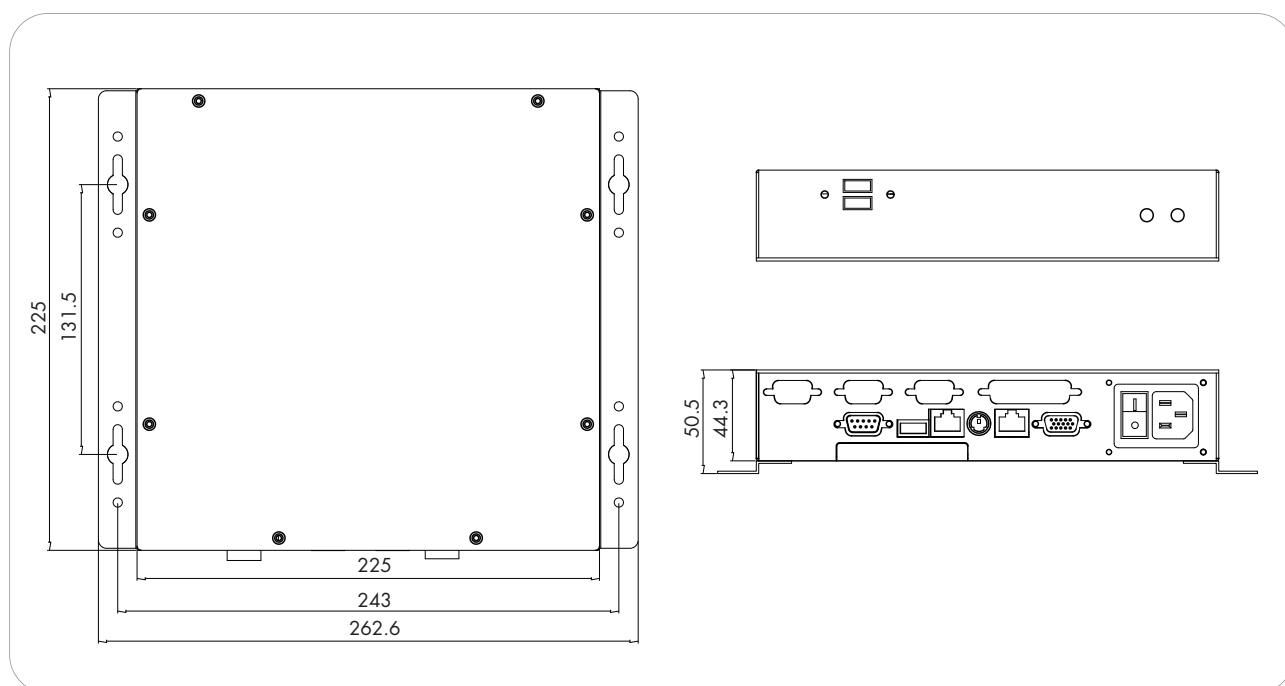
Type	Embedded PC
Operating system	Windows XP embedded
Operation	Fan-less
Processor	AMD LX800 500MHz
Chipset	AMD LX+CS5536+ITE8888G
BIOS	Phoenix-Award 4Mbit with RPL/PXE LAN Boot ROM. SmartView and customer CMOS backup
Memory	
DDR	256MB SODIMM
DOM	8GB
Back-up Industrial CompactFlash™	8GB Type II (on request)
Communication ports	
RS485	3 ports for Eos Array bus management 1 port for inverter and energy meters
Ethernet	1 for internet/LAN connection 1 for local access

Other Ports	
USB	3, for local access and external modem connection
LED	
Status and Colours	Green for power-on Orange for DOM memory access
Connections	
RS485	3 pole detachable screw terminal block per port
Ethernet	RJ-45 connector (10/100Base-T)
USB	High speed USB 2.0

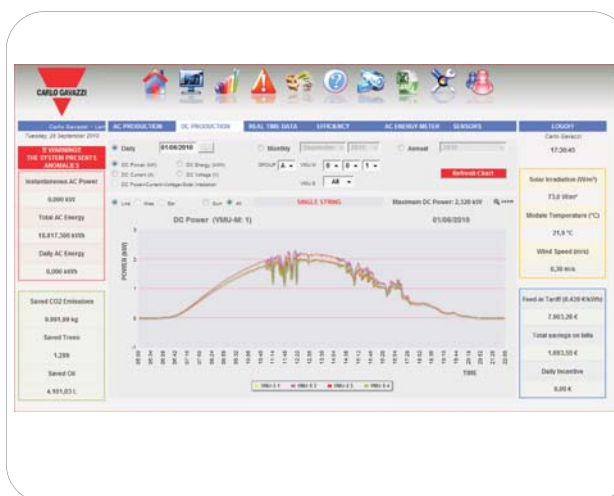
Eos Box - Power Supply Data

Power Supply	100 to 240 VAC
Power Consumption	70 W

Eos Box - Dimensions (mm)



Eos Box - Software



Eos Box software pages available:

- Home page:
Daily AC power of plant.
- Plant page:
Detailed graphs and data table of all available variables.
- Alarms page:
alarms, warnings, events, commands list.
- Economy page:
economical parameters of the installation.
- Information page:
plant description with relevant technical data, financial highlights, energy production data source.
- IP cam page:
live images from the photovoltaic park.
- Monitor page:
combination of main graphs for accurate plant analysis.
- Export page:
data base export in Excel® spreadsheet.
- Setting page:
access to the configuration of all parameters of Eos Box.
- Account page:
access to LAN and Internet configuration.

Eos Box software main functions:

- AC production display.
- DC production display.
- Real time data display.
- Efficiency of the whole photovoltaic park under control (DC and AC part).
- Efficiency of the DC part of the installation.
- Efficiency of the DC to AC conversion inside the inverters.
- BOS: efficiency of the AC part of the photovoltaic park under control.
- Display of: solar irradiation, temperature, wind speed.

Eos ASB

String Box

Eos ASB is a string-box based on combinations of Eos Array modules which perform a complete control of the PV strings being connected. The core unit is VMU-M which performs data and event logging and has the additional task to manage the communication bus among different modules like VMU-S the string controller, VMU-P the environment measuring unit (PV cell temperature, air temperature, sun irradiation and wind speed) and VMU-O the I/O unit. This latter unit, when available, can provide two relay outputs so to manage alarms or/and external loads (like a lighting system; a module washing system and so on).

Component control functions like monitoring of switch disconnector status and correct working of surge arrester are available as standard.

The string-box is supplied completely pre-assembled.

All devices in the string box are already wired in the panel.

Thanks to the modular concept of Carlo Gavazzi solar control solutions, the system integrator can size the PV plant according to his project philosophy and install all equipment in a very easy and fast way.



Perform complete control

• Security and safety features

In addition and according to the needed model, two additional functions are available:

- Antitheft control performed on all modules being connected by means of specific fibre optic sensors (from one two three according the dimension of the area being controlled by the string box);
- Block diodes on single string in case of application needs.

• Strings control

Eos ASB can manage up to 250 A total strings current with standard string protection.

• Chassis

The string box enclosure is available in two sizes and is made of a special thermoplastic material which grants an IP66 protection degree with proper safety and long life characteristics.

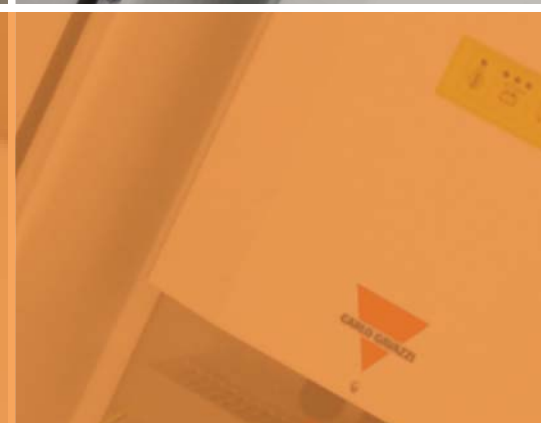
General Data

String rated inputs	
Current	12 A @ 60°C
Voltage	1000 VDC
String measurement inputs	
Current	±(0.5% RDG+2 DGT) from 0.05 A to 12 A
Voltage	±(0.5% RDG+2 DGT) from 20 V to 1000 V
Power	±(1% RDG+ 2DGT)
Energy	±(1% RDG)
Protection	
Fuse holder	Integrated into the string module
Fuse size	10.3x38 mm (IEC269-2-1)
Temperature probe inputs	
Type	2 inputs (PV module; air) Pt100, Pt1000
Irradiation sensor input	
Range	1 input 0 to 120 mVDC
Accuracy	±(0.1% RDG+1DGT) 25% to 120% FS.
Wind speed sensor inputs	
Range	1 input 0 to 1000 Hz max, duty cycle 50%
Accuracy	±(0.01% RDG+1DGT) 25% to 110% FS.
Digital inputs	
Purpose	2 inputs Detection of the status (ON/OFF) of the switch disconnecter, Detection of the status of the surge arrester
Digital output	
Purpose	2 outputs Alarm notification as a String alarm or other alarms
Data logging	
Stored data type	Variables: V, A, W, Wh, PV module temperature, air temperature, irradiation, wind speed and string efficiency
Storage interval	Selectable: 1-5-10-15-30-60 minutes

Sampling time	2 s
Storage method	Circular FIFO
Memory type	Flash
Memory retention time	10 years
Events logging	
Type of stored events	Digital output status change (real and virtual alarms), switch disconnecter status, string alarms (see "String control") and theft alarm, surge arrester working status. The events are recorded as soon as they occur.
Number of events	Max. 10.000
Storage method	Circular FIFO
Memory type	Flash
Memory retention time	10 years
String control	
Function selection	Match max. control or median control
String window alarm	A window alarm can be set as a string power control
Other variable alarms	The alarms can be connected also to the string current and voltage.
"PV string" efficiency measurement	
Antitheft control	Three type of controls are available
Other controls	Antitheft control on PV modules sensed by means of fibre optic technology
Communication port	Fuse blow detection and wrong PV module connection
Purpose	RS485 or fibre optic according to the model
Purpose	Transmission of all data and alarms All parameter can be programmed using freeware EosArraySoft
Power supply (H)	
	90 to 260 VAC

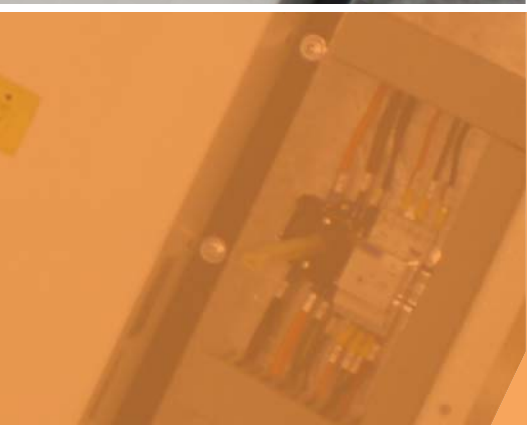
Available models with "S" standard protection (up to 250 A)

Model	Eos ASB											
Strings	10	10	12	12	12	12	12	12	12	12	15	13
Power Supply	H	H	H	H	H	H	H	H	H	H	H	H
String fuse (A)	12	12	12	12	12	12	12	12	12	12	12	12
String voltage (kV)	1	1	1	1	1	1	1	1	1	1	1	1
Protection	S	S	S	S	S	S	S	S	S	S	S	S
Environ. measures	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Digital I/Os	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Com. port	RS485	RS485	RS485	RS485	RS485	RS485	RS485	RS485	FO	FO	RS485	RS485
Surge arrester	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Antitheft channels	No	No	No	No	2	2	2	2	2	2	3	3
Block diode	No	No	No	No	No	No	Yes	Yes	No	No	No	No
Switch-board size	A	A	A	A	A	A	B	B	A	A	B	B



Wind Energy

EcoEnergy
equipment



Presentation	2
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Wind Turbine	
WT3	32
Wind Management Cabinets	
WTC 3	34
Wind Battery Charger and Cabinet	
WTC 3BC	36
Complementary	38

Aeolic energy is the result of the conversion of the kinetic energy of the wind into other forms of energy, either electrical or mechanical. Nowadays, it is mostly converted into electrical energy by means of wind turbines. Aside from pumps for irrigation or drainage, windmills were mostly used to grind grain. Thus, we still speak of "windmills" today, even when we are talking about machines that do not actually grind, such as sawmills and hammer mills.

Today, wind generation is the fastest growing energy production source in the world, with a total power of over 158 GW installed by the end of 2009, and over 38 GW of newly installed power in the same year (Source GWEC 2010 and WWEA 2010). All the wind turbines installed generate a total of 340 TWh, equivalent to the total energy demand of a country like Italy, and equalling 2% of global energy consumption. The capacity of wind power installed

in the world has grown by 170% in just five years and it has become the primary source for new power generation installations, with a share of 39%.

In Europe in 2009, 10.5 GW were installed, of which 582 MW offshore, taking the total wind power capacity up to 76.2 GW.

The forecast is that in 2014 global wind capacity will stand at 409 GW, up from 158 GW at the end of 2009. By 2014, 62.5 GW of new capacity will be added to the global total, compared to 38 GW in 2009 (Source GWEC 2010). The outlook is that by 2020 wind generation will contribute at least 12% to global electricity consumption with 1500 GW produced globally (Source: WWEA 2008).



Wind Energy

Micro, mini and aeolic

Wind generating systems can be divided into 3 main power divisions: microaeolic below 20 kW, miniaeolic from 20 kW to 200 kW, and aeolic above 200 kW. Today the biggest generators can provide up to 7 MW and have rotors of 126 m in diameter. However 10 MW offshore generators are already under development.

Horizontal and vertical turbines

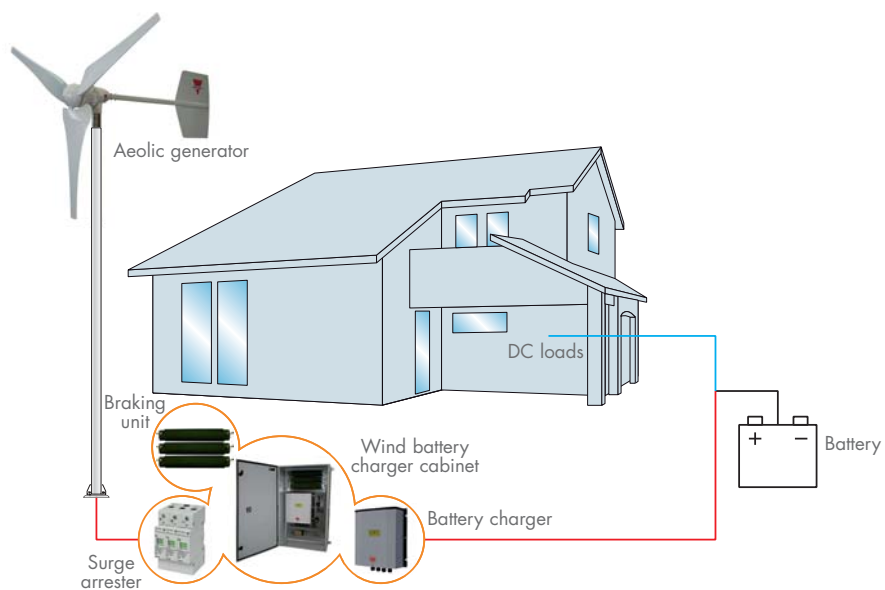
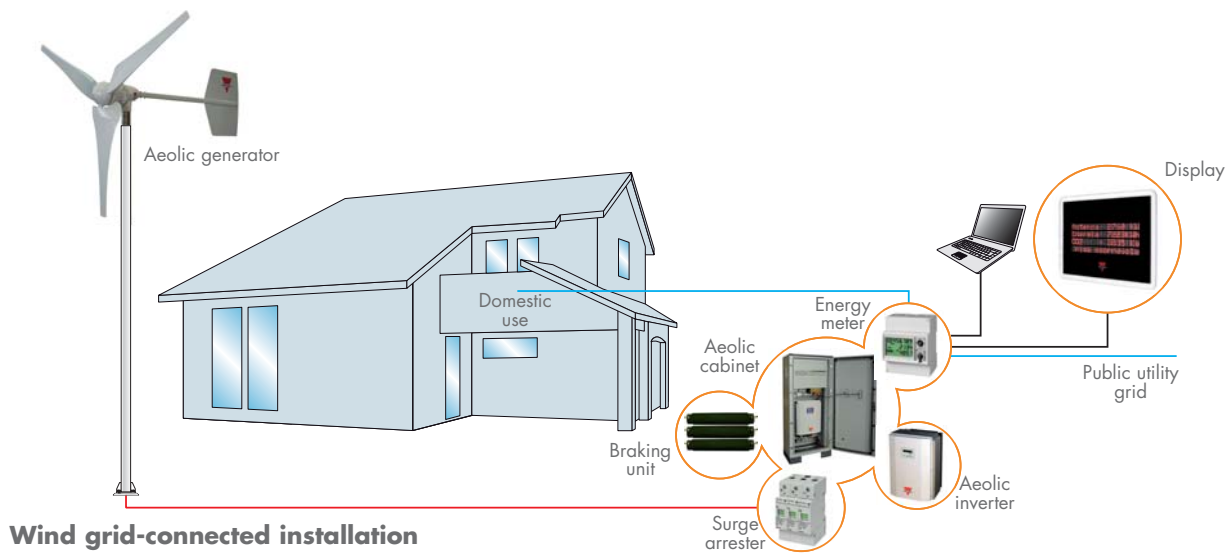
The global energy generated by the aeolic system is made by horizontal axis wind turbines (HAWT) but in microaeolic system, the turbines can also be of vertical axis type (VAWT). Carlo Gavazzi is concentrating its efforts on the HAWT both in the micro and mini aeolic systems.

Grid connected and off grid

Whilst the big aeolic generators are installed in aeolic parks, are connected to the public grid and become power stations for a certain area, micro and mini aeolic turbines are suitable for domestic, business, industrial and farm use, also in remote areas, and can be either grid-connected or off grid.

Carlo Gavazzi turnkey systems

Carlo Gavazzi provides a hassle free microaeolic solution, with all the required devices for the installation integrated into specific cabinets, so that the installer can carry out as few operations as possible. All that remains to be done is the foundation for the wind turbine and the mechanical assembly.



Wind off grid installation

WT 3

Mistral3K - Horizontal wind turbine

The Mistral3K micro wind turbine has been designed according to state-of-the-art technology for Horizontal Axis Wind Turbines (HAWT).

This turbine delivers high efficiency energy with a low level audible noise.

The main concept of this turbine is simplicity. There are as few components as possible for easy installation and commissioning, limiting weight and enhancing long term reliability.

The generator is directly mounted onto the propeller shaft, held in place by high performance ball bearings, without any gears or brakes. The blades have an exclusive profile with a fixed pitch which, together with a brand new generation of lightweight and resistant materials, provides high performance, very low inertia and limited mechanical losses. In the best conditions, this microturbine can have a total efficiency of up to 39%.

Electric power transmission between the moving parts and the fixed parts of the system is guaranteed by means of slip rings that allow total rotation of the wind turbine on the pole.



Harness power from the wind

• Gearless

No mechanical losses due to unnecessary additional reductions that typically have a limited efficiency.

• Generator

The generator is an 18-pole permanent magnet generator with 3 phase output from 50 to 240 VAC.

• 3 kW maximum output power

The maximum output power of the turbine is 3 kW at the maximum operating speed of 13.5m/s, although the rated output power is 2 kW measured at 11 m/s according to the related norm IEC61400-2.

• Furling protection

The turbine is protected against strong winds by passive furling protection. When the wind speed reaches around 13 m/s the tail of the turbine furls, moving it transversally. The generator still rotates and generates energy, but much less. If the wind speed returns to below the limit, the tail goes back to a straight position, the turbine faces the wind direction again and normal operation is restored.

• Low cut-in speed

The turbine starts to rotate at 2.5m/s of wind speed. The power generation starts at 3m/s.

General Data

Type	18 poles Permanent Magnet Generator
Rated output power	2.0 kW @ 11.0 m/s (Acc. to IEC61400-2)
Maximum output power	3.0 kW
Minimum voltage	50 Vrms @ 145 rpm
Maximum voltage	240 Vrms @ 780 rpm
Insulation Class	F

Mechanical Data

Propeller diameter	2.49 m
Sweep area	4.87 m ²
Overall weight	53 kg
Blades weight	9.3 kg
Tail weight	9.6 kg
Body+generator weight	34 kg
Propeller blades	3
Propeller blades material	Composite carbon fibre
Stand material	Zinc-coated steel
Rotor blades protection	Anti-UV, chemical resistant, temperature resistant
Colour	RAL9010

Turbine Braking Control

Automatic system	1) Furling protection begins at 13.5m/s wind speed 2) Over speed electromagnetic brake using WT13 inverter and WTB3 braking box
Manual	Short circuit braking and resistor braking using WTB3 braking box

Turbine Data

Type	HAWT Three Blades Propeller Turbine
Speed range	0 rpm to 800 rpm
Operating speed rotation	145 rpm to 780 rpm
Rated speed	690 rpm @ 12 m/s
Start-up wind speed	~2.5 m/s
Cut-in wind speed	3.0 m/s
"Furl Protection" wind speed	Start @ 13.5 m/s
Survival wind speed	≥50 m/s

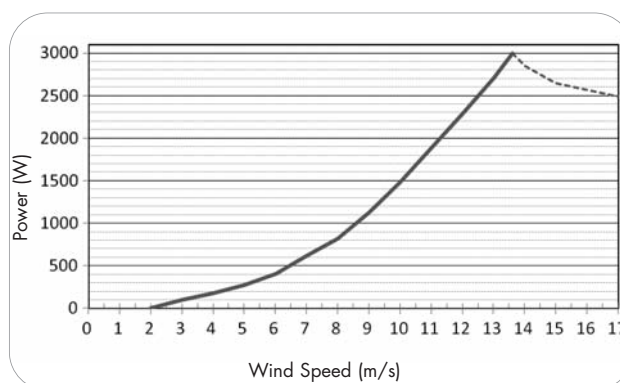
Environmental Data

Ambient temperature	-20 to +50°C
Noise level	< 60dB (Acc. to IEC61400-11)
Humidity	0 to 95% RH
Corrosion protection	Protective paint
Protection degree	IP54

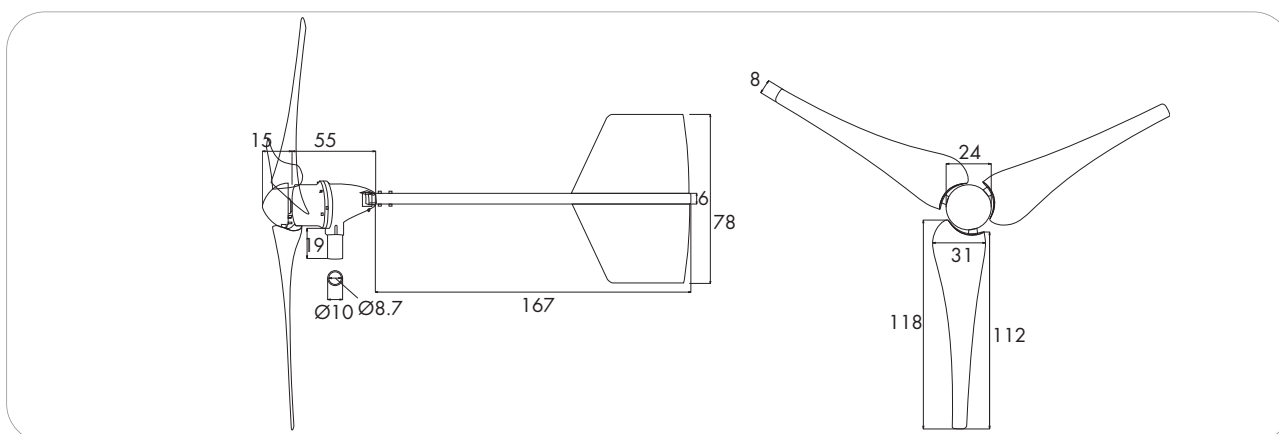
Standard Norms and Certifications

According to	IEC61400-2
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Power Curve



Dimensions (cm)



WTC 3

Mistral3k management cabinets

WTB 3 and WTC 3 are products designed by Carlo Gavazzi for the Mistral3k wind turbine remote management. They both manage the wind generator braking functions. There are 2 modes: resistor braking and total stop.

The resistor braking slows down the turbine in strong wind conditions, the total stop blocks the turbine when the strong wind conditions become persistent.

The braking functions can be overridden by front panel switches when it is necessary to manually stop the wind generator.

The WTB 3 cabinet is merely a braking function cabinet, with the option of having surge protecting devices to increase product safety, whilst the WTC 3 is a complete in-grid solution. The single phase grid-connected inverter is integrated into the cabinet. With the WTC 3 cabinet all that is left to do for the wind system installation is the connection

of the wind turbine and the grid to the cabinet terminals. Both cabinets can be equipped with an energy meter in order to display, locally or on a remote device, the power produced, the energy, the voltages and the currents.



Rely on "plug-and-play" equipment

• Easy installation

WTB 3 has to be connected to an external device capable of managing the power produced by the wind turbine, either to feed the energy into the grid or to charge batteries. Apart from this, the connection to the turbine is simple and fast, with only four terminals.

WTC 3 provides the easiest system when the power provided by the wind generator needs to be fed into the grid. There are just 4 wires to be connected to the generator and 3 wires to the grid.

• Ready configured inverter

The inverter built into the WTC 3 cabinet is ready configured in order to perfectly match the Mistral3k wind turbine. There is no need for further tuning.

The inverter can be programmed according to different national DNO recommendations: VDE0126 or DK5940.

• Strong wind safe operation

The monitoring software detects dangerous wind conditions through the voltage generated by the turbine and activates the soft braking first. If wind conditions remain dangerous for several minutes, the turbine is totally stopped. Every 30 minutes the conditions are checked in order to restore normal operation when wind strength becomes safe.

• Energy monitoring

Both devices can be equipped with a Carlo Gavazzi EM24 energy meter. This unit is equipped with an LCD display for direct reading of produced power, energy to the grid, voltage, current, frequency, yield and saved CO₂.

The built-in RS485 line allows data reading by remote PCs with Carlo Gavazzi PowerSoft, or with Carlo Gavazzi DPY displays.

The EM24 outputs can be used to drive.

Input Data

AC connection	3-phase
Voltage range	0 to 240 VAC
Operating voltage range	30 to 200 VAC
Max. current	18 A

Output Data

WT 3 AC connection	1-phase
Nominal power	3 kW
Max. power	3.3 kW
AC voltage	230 VAC $\pm 5\%$
Frequency	50 Hz
Max. current	14 A
Interface protection	VDE / DK5

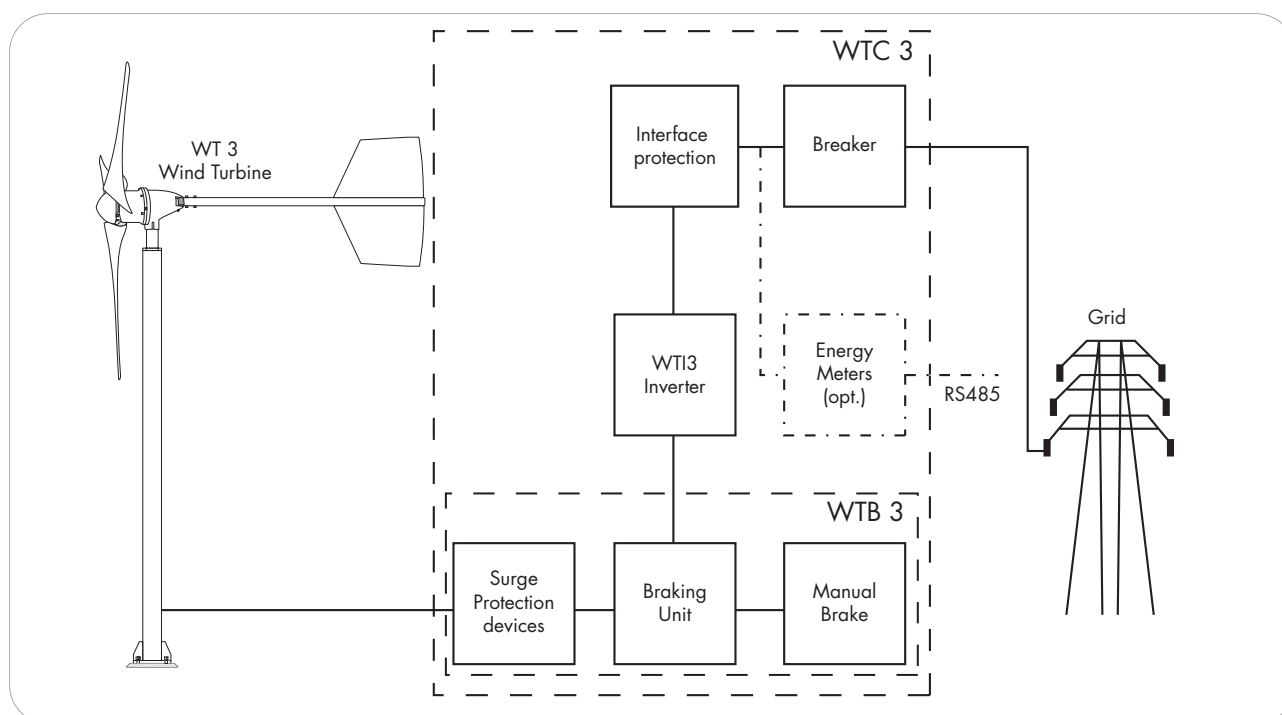
Environmental Data

Operating temperature	-10 to +40°C
Storage temperature	-25 to +80°C
Cooling	Forced ventilation
Humidity	0 to 95% (non condensing)
Installation location	Outdoor/Indoor
Protection degree	IP65

Mechanical Data

Dimensions (HxWxD)	140x60x45 mm
Weight	140.0 kg

Block diagram



WTC 3BC

Wind battery charging

The WTC 3BC is a product designed by Carlo Gavazzi for battery charging from wind energy. Inside the WTC 3BC cabinet are all the devices that are normally required in order to convert the wind turbine energy into battery charging, manage the wind turbine and protecting it against harsh wind conditions. It is also IP65 and can thus be installed outdoors with specific protections.

The inverter of the battery charger allows charge up to 2.4 kW, using a microcontroller for digital accuracy and full automatic operation. The input section of this battery charger is very similar to the wind turbine inverter, with an MPP tracker, designed in order to achieve the best ratio between voltage and current according to the actual turbine performance. The power curve of the Carlo Gavazzi Mistral3k is already stored in the battery charger providing the highest output in any wind condition. The charger provides two relay outputs in order to manage the "soft" and

"hard" braking of the wind turbine when wind conditions become extreme and could compromise the wind turbine survival. The battery charger is integrated into Carlo Gavazzi WTC 3BC and the braking function is managed internally in the cabinet and does not require any external device.



Convert wind power into electricity

• High charging power

The WTC3 BC battery charger is a very high power charger, up to 2.4 kW. The battery charging current can be set, by the user, up to 50 A. The only available battery voltage is 48 V in order to comply with European low voltage directives.

• Battery protection

If the battery charge level is above the float level "soft" braking is activated so that the turbine will slow down and limit charging current.

In case of battery overcharge, the charging will shut down and switch to "float" for 10', then go back to charging for 1'. This behaviour prevents battery life being shortened and early failures.

• Input overvoltage protection

The battery charging unit is equipped with input over voltage protection that prevents high voltage transients getting into the unit. Besides this, in order to increase the overall safety of the system, the cabinet could be equipped, upon request, with built in surge protecting devices.

• High wind protection

When the wind conditions become severe, for the turbine survival, "soft" braking is actuated, by inserting resistors in parallel to the turbine: this action will increase the mechanical resistance of the generator and slow down the rotation.

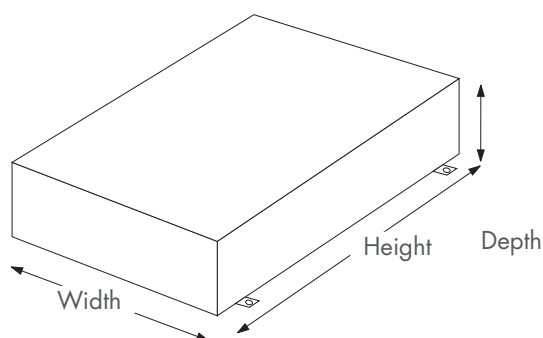
If the high wind conditions become persistent "hard" braking is actuated by shorting the wind turbine terminals and consequently stopping the rotation of the blades.

Wind conditions are tested every 30 minutes and if safe wind speed is verified the brakes are released and normal operation is restored.

Technical Data

Max. output power	3200 W	Battery low protection voltage	44 VDC ± 1 V
Rated output power	3000 W	Float voltage	58 V ± 1 V, 1 min charging per 10 min
Input voltage range (line voltage)	50~200 Vrms	Max. charge current	adjustable
Rated input current	20 A	Working temperature	-40°C to +40°C
Max. charger efficiency	>90%	Installation location	Outdoor/Indoor
MPPT efficiency	>95%	Degree of protection	IP54
Stand-by power	<5 W	Weight Kg/Lb	60 kg/132.28 lb
Recommended 48 V rechargeable batteries	200AH/350AH/500AH/600AH		

Dimensions



H x W x D mm/inches	1000x600x300 / 39.37x23.62x11.81"
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Complementary

EcoEnergy
equipment

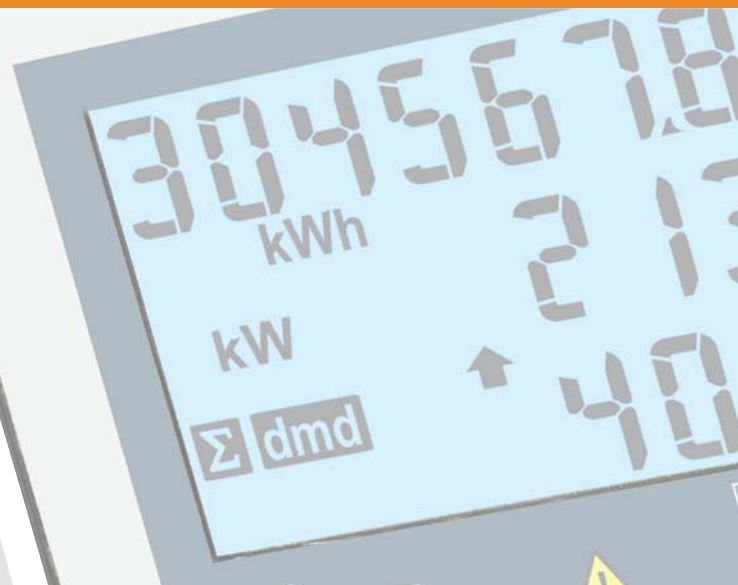


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EM

Energy meters

The metering of the produced energy (provided to the grid or locally consumed) is of fundamental importance in any renewable energy generation system, first of all because the economic incentives strictly depend on the measured value. Carlo Gavazzi is able to supply a complete range of energy meters and analyzers that perfectly fit the different application needs. All these meters can be supplied with MID certification (Annex D). As a result of Carlo Gavazzi innovation, the EM-series features a number of benefits both for the installers and the end-user: a wide and bright display, compact size, easy connection and easy programming, also by means of dedicated software. Accuracy, reliability, MID certification, application analysis and style are the key words behind the success of the Carlo Gavazzi EM-series.



Meter the power fed into the grid

EM24, EM26

- Tamper proof and revenue approval for billing purpose. Tamper proof capability using proper covers, thus protecting all input/output screw terminals. Lockable programming access using a password and a seal on the front selector.
- Easy variable scrolling, by means of the front joystick. Time saving installation system using self-power supply and automatic phase detection.
- Direct variable page access by means of the front four position selectors programmable by the user.
- Application orientated programming structure. Selection of eight different applications: Basic domestic - Shopping malls - Advanced domestic - Multi domestic (also camping and marinas) - Solar - Industrial - Advanced industrial - Advanced industrial for power generation, providing only the required programming parameters and the display variables, thus simplifying the installation and the display readout.
- 65 A direct connection available (EM24 AV9/2 only). Connections for cables with cross-section area from 6 to 35mm² instead of by-passing types assuring a "contactor type" wiring and connection protection.

- Programming software tools allowing the user to configure the instruments using the PC and the RS485, the RS232 or ports, making the programming job easier and faster.

EM11

- Power analyzer features in an ultra compact energy meter. New solution providing full parameter control, also saving space in the small switch panels. Relay output to switch OFF a non priority load, thus preventing a line overload condition.

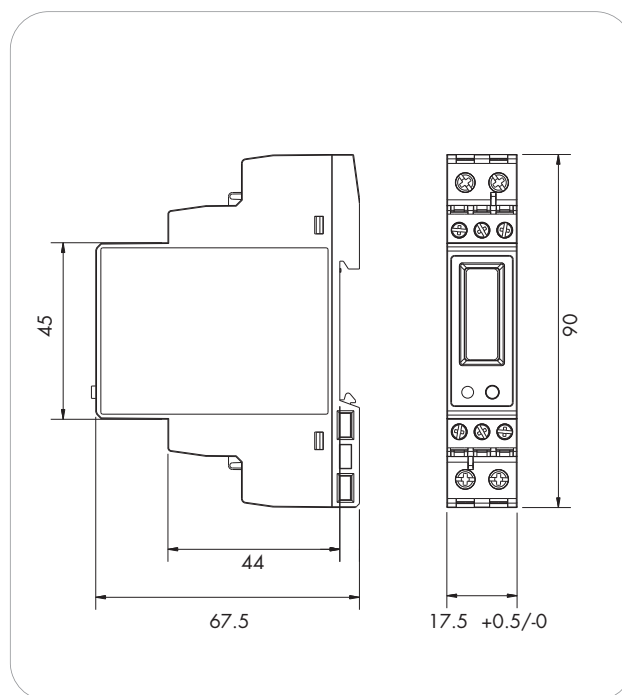
EM21 72D

- One product, 3 different installation modes. Detachable display to carry out up to 3 installation modes from the same instrument: DIN-rail, panel mounting or transducer.
- Time saving installation system exploiting self-power supply, automatic phase detection and application oriented programming procedure.

EM10 - EM11

Description and measurement	EM10: 1-phase energy meter (kWh) EM11: 1-phase energy analyzer V_{IN} , A, Hz, W, W_{dmd} , var, PF, kWh, kvarh. TRMS method
Input Specifications	
Range code	120 VAC 230 VAC Ib: 5 A, I _{max} : 32AAC
Accuracy	EM11: $\pm 0.5\%$ RDG (V, A)
Active energy	Class 1 (EN62053-21)
Reactive energy	(EN50470-3) Class 2 (EN62053-23) (EM11 only)
Display	4 DGT (inst. variables, EM11 only) 5+1 DGT (energies), LCD
Output Specifications	
Out 1 (Pulse)	1-open collector
Out 1 (Alarm)	1-relay (EM11 only)
General Specifications	
Power supply	Self power supply
Approvals/Marks	CE, MID certification

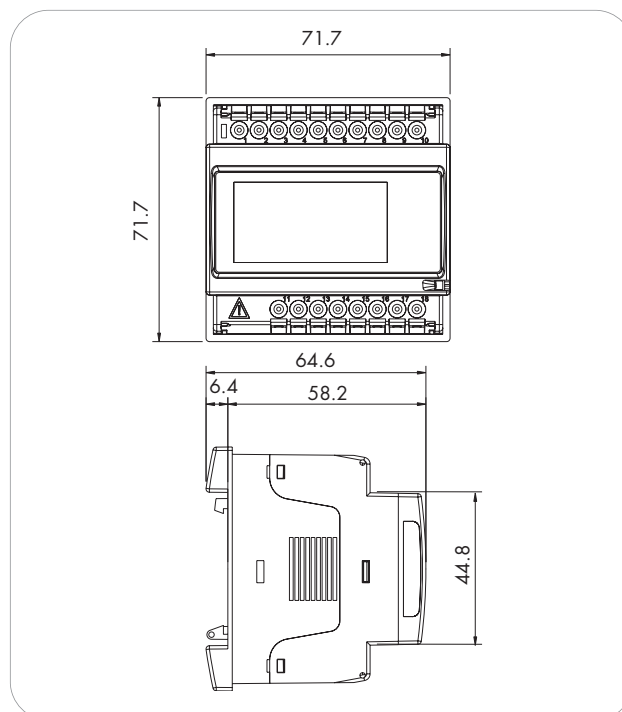
EM10 - EM11 - Dimensions (mm)



EM21 72D

Description and measurement	3-phase energy analyzer with dual mounting capability: panel and DIN rail. W, var, PF, Phase-sequence, V_{LL} , V_{LN} , A TRMS method
Input Specifications	
Range code	120/230 VAC, 400 VAC I _n : 5 A; I _{max} : 6 A
Active energy	Class 1 (EN62053-21)
Reactive energy	Class B (EN50470-3) Class 2 (EN62053-23)
Display	3 DGT (inst. variables) 6+1 DGT (energies), LCD
Output Specifications	
Out 1 (Pulse)	1 static opto-mosfet
Out 2 (Serial port)	RS485 (2-wire, Modbus) M-BUS by means of VMU-B
General Specifications	
Power supply	Self power supply
Approvals/Marks	CE

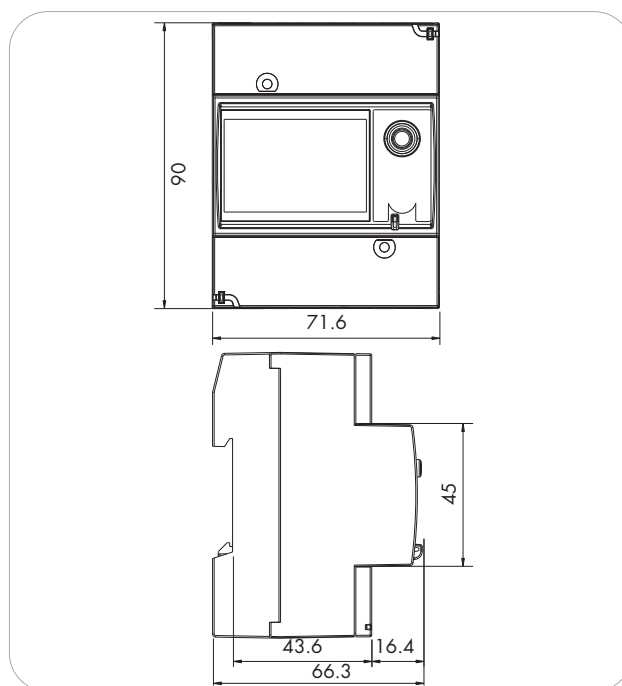
EM21 72D - Dimensions (mm)



EM23 DIN

Description and measurement	3-phase energy analyzer W, var, A, kWh, kvarh TRMS method
Input Specifications	
Range code	400 VLLAC Ib: 10 A; I _{max} 65 A
Active energy	Class 1 (EN62053-21)
Reactive energy	Class B (EN50470-3) Class 2 (EN62053-23)
Display	3x3 DGT (inst. variables) 6+1 DGT (energies), LCD
Output Specifications	
Out 1 (Pulse)	1 static collector
General Specifications	
Power supply	Self power supply
Approvals/Marks	CE, MID certification

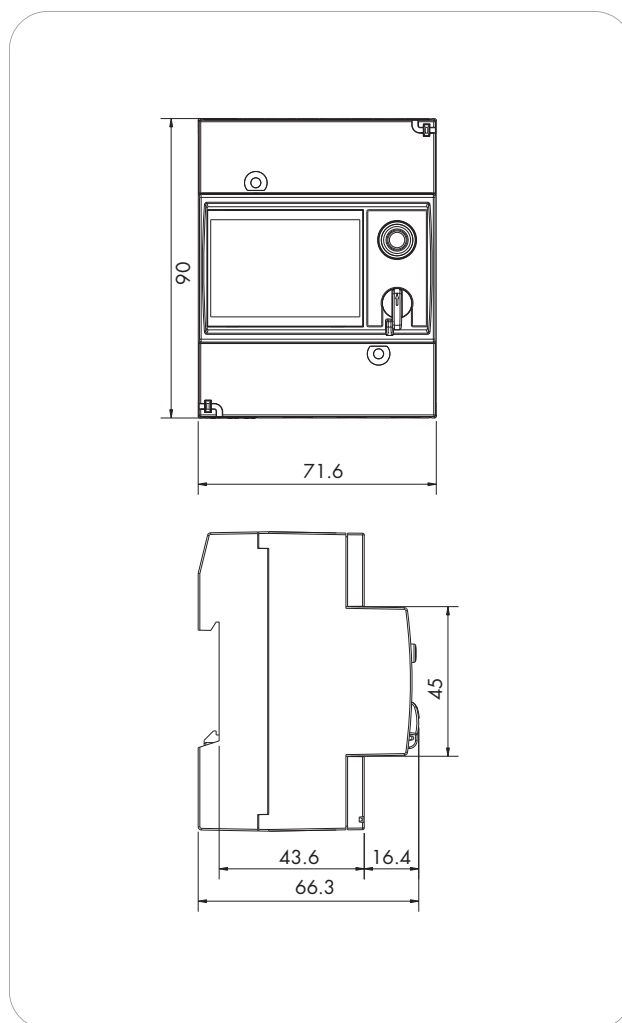
EM23 DIN - Dimensions (mm)



EM24 DIN

Description and measurement	3-phase energy analyzer Sys: V _{LL} , V _{LN} , var, VA, W _{dmd} , W, VA _{dmd} , Hz, kWh, kvarh, hour counter, gas and water Max: A _{dmd} , W _{dmd} , VA _{dmd} . Single-phase: V _{LL} , V _{LN} , A, W, var, VA, PF, A _{dmd} , kWh, kvarh. TRMS method
Input Specifications	
Range code	120/208 V _{LL} ; 400 V _{LL} In: 1/5 A, I _{max} : 10AAC. 120/208 V _{LL} , 230 V _{LL} , 400 V _{LL} Ib: 10 A, I _{max} : 64AAC
Accuracy	±0.5% RDG (V, A)
Active energy	Class 1 (EN62053-21)
Reactive energy	Class B (EN50470-3) Class 2 (EN62053-23)
Display	3x4 DGT (inst. variables) 8 DGT (energies), LCD
Output Specifications	
Out 1 (Pulse)	2 open collector/relay
Out 1 (Alarm)	2 open collector/relay
Out 2 (Serial port)	RS485 (2-wire) M-BUS by means of VMU-B
Inputs	3 digital inputs
General Specifications	
Power supply	Self power supply Auxiliary power supply: 18 to 60 VAC/DC, 115/230 VAC
Approvals/Marks	CE, MID certification

EM24 DIN - Dimensions (mm)



EM26 96

Description and measurement

3-phase energy analyzer
 Sys: V_{LL} , V_{LN} , A_n , var, VA, W, W_{dmd} , VA_{dmd} , VA, Hz, %THD-V, %THD-A, kWh, kvarh, hour counter, gas and water
 Max: A_{dmd} , W_{dmd} , VA_{dmd} .
 Single-phase: V_{LL} , V_{LN} , A, W, var, VA, PF, A_{dmd} , kWh, kvarh.
 TRMS method

Input Specifications

Range code 120/208 V_{LL} ;
 400 V/600 V_{LL}
 In: 1/5 A, I_{max}: 10AAC

Accuracy $\pm 0.5\%$ RDG (V, A)

Active energy Class 1 (EN 62053-21)

Reactive energy Class 2 (EN 62053-23)

Display 3x4 DGT (inst. variables)
 8 DGT (energies), LCD

Output Specifications

Serial communication port RS485 (2-wire)

Alarm port 2-relay/open collector

Pulse output 3-open collector/relay

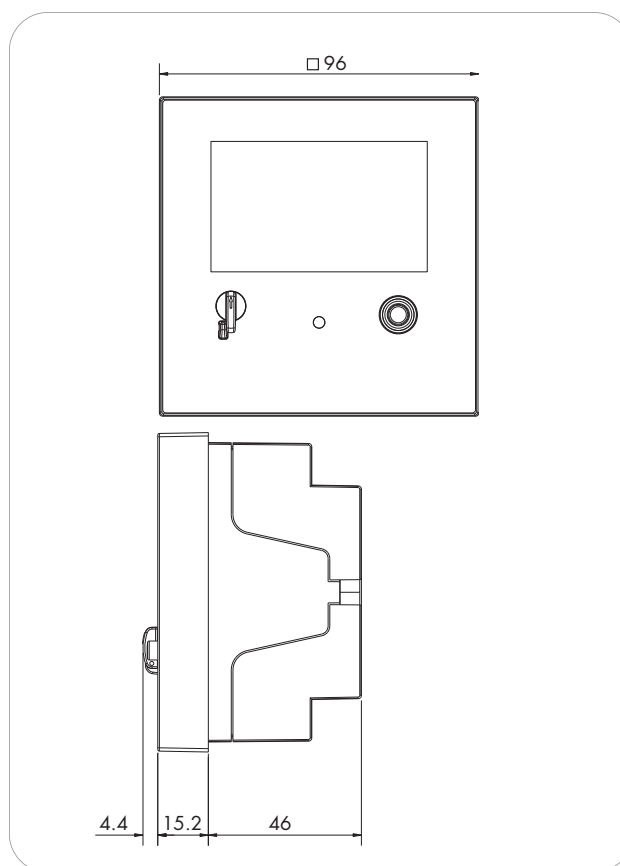
Digital input 3 digital inputs

General Specifications

Power supply 18 to 60 VAC/DC,
 90/260 VAC/DC

Approvals/Marks CE, cULus (only 90/260 V version)

EM26 96 - Dimensions (mm)



Functions and Features

	EM10 DIN	EM11 DIN	EM21 72D	EM23 DIN	EM24 DIN	EM26 96
AC energy metering	Yes	Yes	Yes	Yes	Yes	Yes
Gas and water metering					Yes	Yes
Variables analysis		Yes	Yes		Yes	Yes
Power quality analysis						Yes
CT connections			Yes		Yes	Yes
Direct high-current connections	Yes	Yes		Yes	Yes	
Digital inputs					Yes	Yes
Pulse output	Yes	Yes	Yes	Yes	Yes	Yes
Alarm outputs		Yes			Yes	Yes
Serial communication			Yes		Yes	Yes
Dupline					Yes	

DPC72

Interface protection

When dealing with energy generators connected to the public grid, an interface protective device which monitors both voltage and frequency is required. It has to comply with National regulations or to specific ones defined by each single energy supplier.

If any fault occurs, DPC72, disconnects the generator from the grid, prevents the former from continuing to supply the mains and therefore immediately removes any dangers from both users and devices.

DPC72 B002 is the German version (certified by TÜV according to VDE0126-1-1) of the interface protection for the renewable energy market, which adds to the mandatory requirements other interesting and useful features like the LCD display, an event data logger and counter, and an RS485 communication port allowing the device to be easily programmed and the system to be real-time monitored.

The DPC72 B003 is certified according to the Italian ENEL

DK5940.

Complete in a very compact housing (only 4-DIN modules wide) and with a simplified programming procedure, thanks to the display, the front joystick and the front selector or, as an alternative, by the DpcSoft dedicated software which is downloadable for free from our website.



Monitor electric parameters

• Protection

- Voltage, frequency, phase loss and sequence control with 50ms reaction time for an immediate disconnection of the power generator from the grid in case of fault.
- Tamper-proof features to avoid undesired modification of the programmed parameters.

• Communication and data logger

- RS485 communication port for transmission to PC of the status and the instantaneous and logged data.
- Event counter and data logger for off-line data analysis.
- Multifunction monitoring relay including display, communication port and datalogger in a very compact size (4 DIN modules) for space saving also in small cabinets.

DPC72 B002 - Input Data

Nominal voltage	400 VAC L-L, 230 VAC L-N
Voltage setpoints	
Lower setpoint	
Range	320 to 400 VAC L-L 185 to 230 VAC L-N
Step adjustment	1 VAC
Preset value	320 VAC L-L, 185 VAC L-N
Upper setpoints	
Range	400 to 480 VAC L-L 230 to 277 VAC L-N
Step adjustment	1 VAC
Preset value	460 VAC L-L, 265 VAC L-N
Hysteresis	12 V
Voltage quality setpoints	Floating average
Upper setpoints	
Range	440 to 460 VAC L-L 254 to 265 VAC L-N
Step adjustment	1 VAC
Preset value	440 VAC L-L, 254 VAC L-N
Lower setpoint	None
Integration time	
Range	1 to 30 min
Step adjustment	1 min
Preset value	10 min
System frequency	50 Hz, 60 Hz
Frequency setpoints	
Lower setpoint	
Range	45 to 65 Hz
Step adjustment	0.1 Hz
Preset value	47.5 Hz
Upper setpoints	
Range	45 to 65 Hz
Step adjustment	0.1 Hz
Preset value	50.2 Hz
Hysteresis	0.1 Hz

Timings	
Power ON delay	
Range	1 to 6 s
Step adjustment	1 s
Preset value	1 s
Power ON connection time	30 s, fixed
Upper and lower voltage delay on alarm	0.05 s to 1 s
Upper and lower frequency delay on alarm	0.05 s to 1 s
Voltage quality	<0.05 s, fixed
Incorrect phase sequence or total phase loss	
Alarm ON delay	<50 ms $\pm$ 15 ms (if the monitoring is enabled)
Recovery time	
Upper/lower voltage and frequency	
Setpoints exceeding: <3 s	
Range	0 to 30 s
Step adjustment	1 s
Preset value	5 s
Setpoint exceeding: >3 s	
Range	0 to 180 s
Step adjustment	1 s
Preset value	30 s
Voltage quality	
Range	0 to 180 s
Step adjustment	1 s
Preset value	30 s

DPC72 B002 - General Data

Accuracy (Display + RS 485)	(@25°C $\pm$ 5°C, R.H. <60%, 45 to 65 Hz)
Voltage	$\pm$ (0.5% RDG + 1DGT)
Frequency	$\pm$ 0.01 Hz (45 to 65 Hz)
Temperature drift	<200ppm/°C
Insulation	
Input to relays output	4 kV (1.2/50 μ s), $\geq$ 2k VAC (rms)
Input to RS485 port	4 kV (1.2/50 μ s), $\geq$ 2k VAC (rms)
RS485 port to relays output	4 kV (1.2/50 μ s), $\geq$ 2k VAC (rms)
Environment	(EN 60529)
Degree of protection	
Front	IP50
Screw terminals	IP20

Pollution degree	3
Operating temperature	
8 A output	-20 to +50°C, R.H. <95%
5 A output	-20 to +60°C, R.H. <95%
Storage temperature	-30 to +80°C, R.H. <95%
Weight	Approx. 300g
Approvals	VDE 0126-1-1
CE marking	Yes
LVD	Acc. to EN 61010-1
EMC	Electromagnetic compatibility
Immunity	Acc. to EN 61000-6-2
Emissions	Acc. to EN 61000-6-3

DPC72 B003 - Input Data

Voltage range	380 VAC to 415 VAC
Voltage setpoints	
Lower setpoint	
Range	320 to 400 VAC
Step adjustment	1 VAC
Preset value	320 VAC
Upper setpoints	
Range	400 to 480 VAC
Step adjustment	1 VAC
Preset value	480 VAC
System frequency	50 Hz
Frequency setpoints	
Lower setpoints	
Range	48.5 to 49.8 Hz
Step adjustment	0.1 Hz
Preset value	49.7 Hz
Upper setpoint	
Range	50.0 to 51.5 Hz
Step adjustment	0.1 Hz
Preset value	50.3 Hz
Frequency derivative setpoints	
Range (absolute value)	01 to 1 Hz/s
Step adjustment	0.1 Hz/s
Preset value	monitoring not enabled
Hysteresis	
Voltage	20 VAC
Frequency	0.1 Hz
Derivative frequency	0.020 Hz

Timings	
Power ON delay (T0)	
Range	1 to 6 s
Step adjustment	1 s
Preset value	1 s
Upper (T1) and lower (T2) voltage delay on alarm	
Range	0.05 s to 1 s
Step adjustment	0.05 s
Preset value	0.05 s
Upper (T3) and lower (T4) voltage delay on alarm	
Range	0.05 s to 1 s
Step adjustment	0.05 s
Preset value	0.05 s
Derivative frequency delay on alarm (T5)	
Range	0.05 s to 1 s
Step adjustment	0.05 s
Preset value	0.05 s (if the monitoring is enabled)
Incorrect phase sequence or total phase loss	
Alarm ON delay	<50 ms ± 15 ms (if the monitoring is enabled, no other adjustment is allowed)

DPC72 B003 - General Data

Accuracy (Display + RS 485)	(@25°C ± 5°C, R.H. <60%, 45 to 65 Hz)
Voltage	± (0.5% RDG + 1DGT)
Frequency	± 0.01 Hz (45 to 65 Hz)
Temperature drift	<200ppm/°C
Insulation	
Input to relays output	4 kV (1.2/50 µs), ≥2k VAC (rms)
Input to RS485 port	4 kV (1.2/50 µs), ≥2k VAC (rms)
RS485 port to relays output	4 kV (1.2/50 µs), ≥2k VAC (rms)
Environment	(EN 60529)
Degree of protection	
Front	IP50
Screw terminals	IP20

Pollution degree	3
Operating temperature	
8 A output	-20 to +50°C, R.H. <95%
5 A output	-20 to +60°C, R.H. <95%
Storage temperature	-30 to +80°C, R.H. <95%
Weight	Approx. 300g
Approvals	ENEL DK5940
CE marking	Yes
LVD	Acc. to EN 61010-1
EMC	Electromagnetic compatibility
Immunity	Acc. to EN 61000-6-2
Emissions	Acc. to EN 61000-6-3

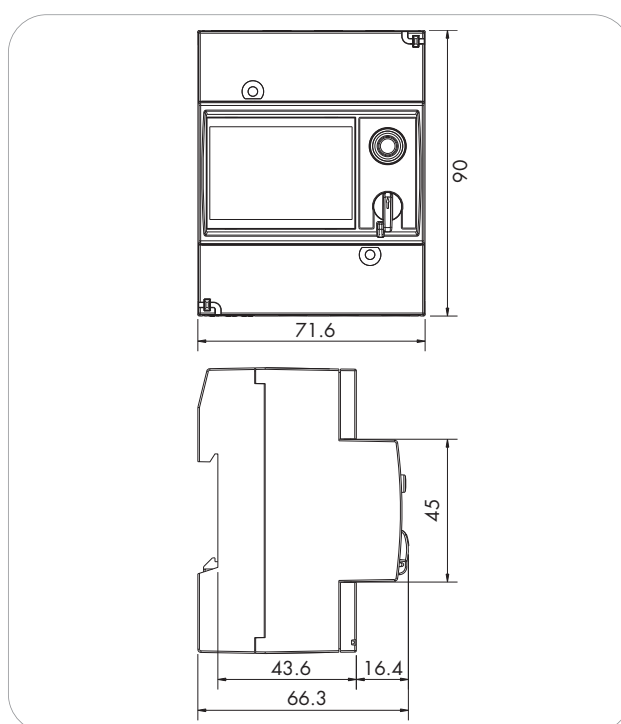
DPC72 B002 & B003 - Output Data

Relay output		1 x DPDT N.E.
Terminals 11, 12, 13 / 8, 9, 10		Voltage/frequency related (both outputs release in case of phase loss or wrong phase)
Relay contact rating (AgSnO ₂)		
Resistive loads	AC1	8 A @ 250 VAC
	DC12	5 A @ 24 VDC
Small inductive loads	AC15	2.5 A @ 250 VAC
	DC13	2.5 A @ 24 VDC
Relay mechanical life		≥ 30 x 10 ⁶ operations
Relay electrical life		≥ 10 ⁵ operations (@ 8 A, 250 V, cosφ = 1)
Relay operational frequency		≤7200 operations/h
RS485		
Type	Multidrop, bidirectional (static and dynamic variables)	
Connections	2-wire (terminals 31, 32, 33)	
Addresses	255, selectable	
Protocol	MODBUS/JBUS (RTU)	
Data (bidirectional)		
Dynamic	Reading only	
Static	Reading/writing	
Data format	1 start bit, 8 data bit, 1 parity bit (even, odd or none (default) control), 1 stop bit	
Speed	9600 (default) or 4800 bit/s, selectable	
Driver input impedance	1/5 unit load, max. 160 devices on the same bus	

DPC72 B002 & B003 - Supply Data

Power supply		Overvoltage cat. III
Rated operational voltage through terminals:	(IEC 60664, IEC 60038) 55, 53, 51	
Delta Voltage	380 to 415 VAC ± 15% 45 to 65 Hz	
Rated operational power		8 VA Supplied by L2 and L3

DPC72 B002 & B003 - Dimensions (mm)



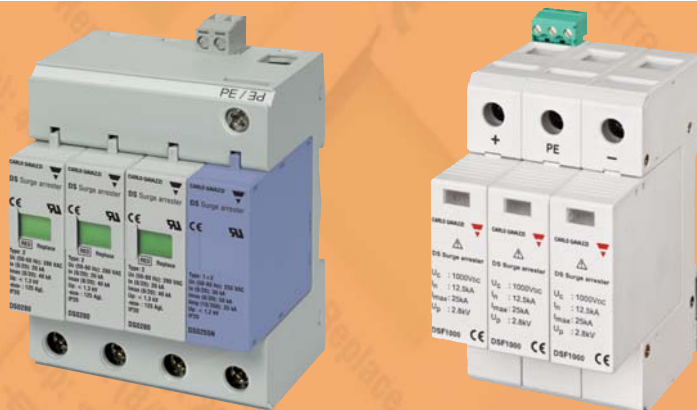
DSA, DSF

Surge arresters

A photovoltaic installation is a long-term investment which needs to be protected from any possible damage from power surges. These surges could affect the AC side, connected to the public grid, or the DC side (from the panel to the string boxes, to the inverter), or the serial communication line which connects the inverter, the measurement and control instruments, the interface protection, and the acquisition system: the damage can be considerable, both in terms of faulty or destroyed devices and in terms of energy production stops, with the consequent money loss.

Voltage surges are peaks induced in the electrical grid because of either low-voltage system switchings or lightning. The former are due to load switching, load variation, switching of producing plants or transformers, and can reach 2 or 3 times the nominal voltage of the system. Inductive or galvanic coupling can give rise to a

surge of more than 2 km around a lightning impact point, reaching up to 20 times the nominal voltage (indirect lightning). All this can cause immediate damage to any electrical device connected to the system where the surge took place, or long-term damage even if the surge values are not very high.



Protect plants from any damages

• AC arrester:

Surge arresters for DIN rail mounting with removable cartridges without the need for a tool, window status indication (green/red), and relevant free of voltage contact. Thanks to the contact, the exhausted removable cartridges can be replaced without leaving the installation unprotected.

Internal wiring and labelled terminals.

Easy installation and maintenance.

• Data line arrester:

Compact arrester for serial communication lines, suitable for any 2-wire bus system, with high lightning protection capability.

The protection of data lines, often omitted, allows complete protection of the inverter, avoiding damage to the control and acquisition data system.

• DC arrester for PV applications:

Surge arresters for DIN-rail mounting, specifically addressed to PV systems. According to the new French standard UTEC 6-740-5 which will be acknowledged as European standard EN50539-11.

Ideal and safest solution for any type of combiner box in PV systems, due to the built-in backup protection which saves cost and space for the no more needed backup fuses and assures protection in case of temporary overvoltages too.

Based on easily removable cartridges, with a red optical indication and voltage free contacts for remote status monitoring. The exhausted cartridges can be replaced without leaving the installation unprotected.

Internal wiring and labelled terminals. Easy installation and maintenance.

Surge Arresters Models

	Code	Uc DC / Uc AC	Description	Acc. to UTE C 61-740-51	Type (EN61643-11)	Remote Contact	Cartridge
DC Surge Arrester for PV systems DSF	DSF52CD600PV	600 VDC	2-pole	Yes	2	Yes	2 x DS0600F
	DSF52XD1000PV	1000 VDC	2-pole	Yes	2	No	2 x DS1000F
	DSF52CD1000PV	1000 VDC	2-pole	Yes	2	Yes	2 x DS1000F
	DSF53XD1200PV	1200 VDC	3-pole (Y)	Yes	2	No	3 x DS1200F
	DSF53CD1200PV	1200 VDC	3-pole (Y)	Yes	2	Yes	3 x DS1200F
AC Surge Arrester DSA_A, DSA_P	DSA52CA280N	280 VAC	1-phase (1+1) for TN and TT systems	n.a.	2	Yes	1 x DS0280 1 x DS0255N
	DSA54CP280N	280 VAC	3-phase (3+1) for TN and TT systems	n.a.	2	Yes	3 x DS0280 1 x DS0255N
Data Surge Arrester DSA_S	DSA51XXS	/	Surge arrester for data transmission systems	n.a.	/	No	/

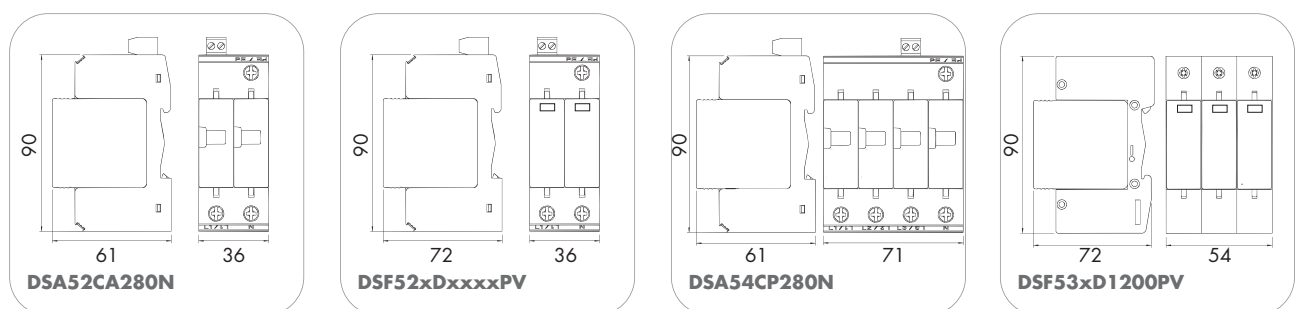
Cartridge Specifications

			DS0600F	DS1000F	DS1200F	DS0280	DS0255N
Max. continuous operating voltage AC	U _c	V	/	/	/	280	255
Max. continuous operating voltage DC	U _c	V	600	1000	1200	350	/
SPD requirement class (EN 61643-11)					Type 2		Type 1+2
SPD requirement class (IEC 61643-1)					Class II		Class I + II
LPZ					1 → 2		0 → 2
Pulsed current (10/350)	I _{imp}	kA	/	/	/	/	25
Nominal discharge surge current (8/20)	I _n	kA	20	12.5	20	20	30
Max. discharge surge current (8/20)	I _{max}	kA	40	25	40	50	
Voltage protection level	U _p	kV	< 2.2	< 2.8	< 4.4	< 1.3	< 1.2
Response time	t _A	ns	< 25	< 25	< 25	< 25	< 100
Temperature range		°C			-40 to 80		

Surge Arrester for Data Specifications

DSA51XXS				DSA51XXS		
Nominal voltage	U _N	V	24	Nominal load current	I _L	A
Max. continuous operating voltage AC	U _c AC	V	19.7	Protection level wire/wire		V
Max. continuous operating voltage DC	U _c DC	V	28	Protection level wire/earth		V
LPZ			0 3	Series resistance	R	Ohm
Transmission frequency	f	MHz	0-100	Response time	t _A	ns
Pulsed current (10/350)	I _{imp}	kA	6	Temperature range		°C
Nominal discharge surge current (8/20)	I _n	kA	20	Protection degree		IP20
				Connections		screwless
				Width	mm	17.5
				Wire section	mm ²	0.14 - 2.5

Dimensions (mm)



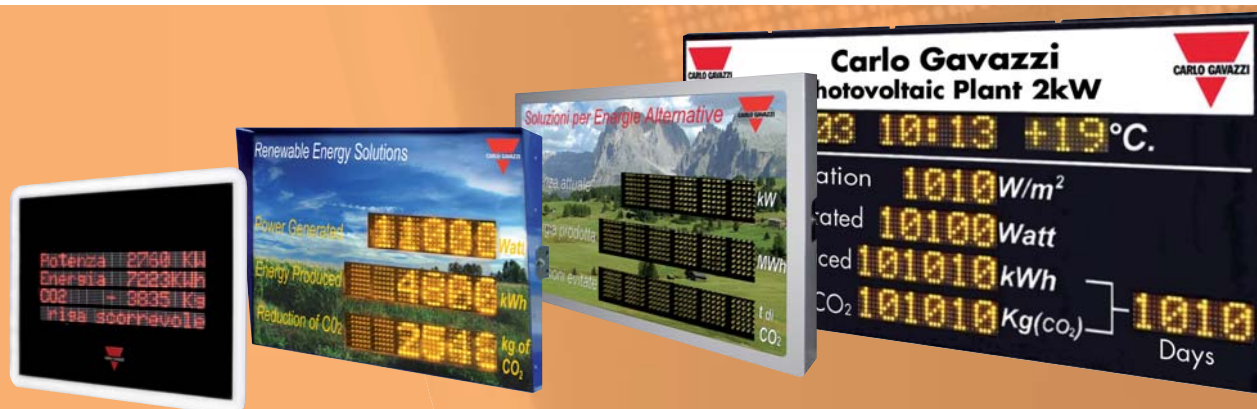
DPY

Digital energy production display

With the DPY displays it is possible to remotely show the production of any power generation system. This device can become a powerful marketing and communication tool. The display communicates, by mean of its serial port RS232/RS485, with Modbus RTU protocol, with the remote device which is actually performing the reading of the data. Carlo Gavazzi suggests using it together with our energy meters EM24. If used in such a way, there is no need for any programming, it is "plug and play".

Standard dimensions are those listed in the data sheet but the display can be tailored to customer requirements, on the DPY 5, up to 1600 x 800mm.

Digits on the DPY 3 and DPY 5 have very bright yellow LEDs, are always 70mm high and can be read from more than 30 m. On the DPY 4 the LEDs are red and they are suitable only for indoor viewing.



Show generated energy and saved emissions

• High visibility

On the DPY 3 and DPY 5: very high brightness yellow LEDs, digit dimension 50x70mm, allowing reading from 30 m and more depending upon light conditions. The brightness of the LEDs is self adjusting, according to environmental light, for power saving in dark conditions.

• Displayable values

The DPY 3 has 3 fields in order to display: actual power, energy produced since installation and CO₂ emissions saved.

The DPY 5 has 5 rows, the first one displaying date/time and external temperature. On this device, besides the readings of the DPY 3, it is also possible to display the Irradiation in W/m² and, on the right hand side of the 'energy since installation' and the CO₂ saved, also the number of days since installation. The DPY 4 has a different concept, as it is programmable. The different "pages" can be programmed individually and can be changed. The last row is a horizontally scrolling message and can also change page by page.

• Indoor/Outdoor

There are 2 constructions, one for indoor installation IP40 and one for outdoor IP54.

• Communication

Communication with the energy meters is by the RS232 or RS485 with Modbus RTU protocol. With the RS485 the maximum distance is 1,200m. Only 2 shielded wires are required.

• Customised look

Carlo Gavazzi provides standard DPY 3 panels with attractive nature panoramas, standard wording and the Carlo Gavazzi logo.

The DPY 4 and DPY 5 standard versions come with a black background and the Carlo Gavazzi logo.

All the displays can be customized with characters, wording, company logo and background pictures provided by the customer.

General Data

DPY 3 Connection type	RS232 / RS485 COM
DPY 4 Connection type	RS232 / RS485 COM
DPY 5 Connection type	RS485 COM
Power supply	230 VAC @ 50 Hz

DPY 3 & DPY 5- Mechanical Data

Housing material	Aluminium
Frontal screen	Anti mirror screen
Frontal layout	Standard or custom (vinyl with 4 colours)
Fixing devices	Wall mounting brackets
Nuts/screws material	Stainless Steel
Installation location	Indoor/outdoor
Protection degree	
Indoor Version	IP40
Outdoor Version	IP54

DPY4 - Mechanical Data

Housing material	Aluminium
Frontal screen	Lexan 8mm/0.31"
Frontal layout	Standard or custom (vinyl with 4 colours)
Fixing devices	Wall mounting brackets
Nuts/screws material	Stainless steel
Installation location	Indoor
Protection degree	IP20

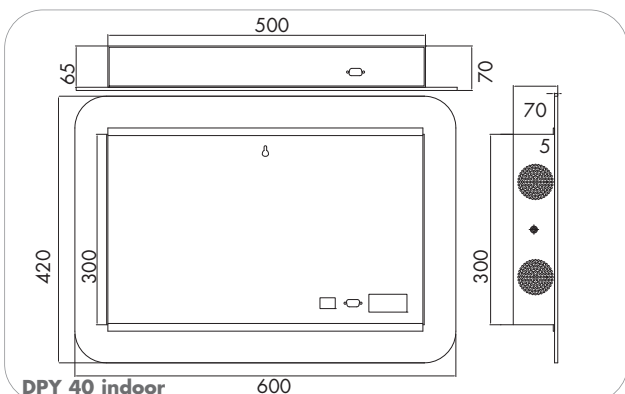
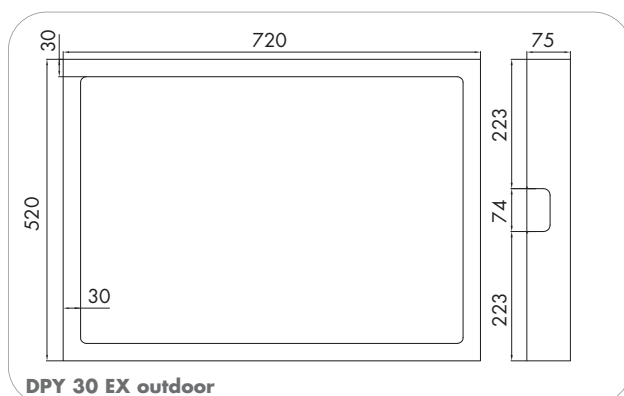
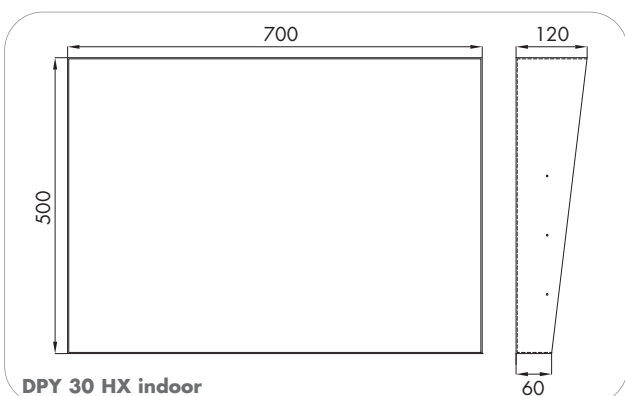
DPY3 & DPY5 - Display Data

DPY 3 Number of lines	3
Digit per line	5 + 6 + 6
DPY 5 Number of lines	5 (+1 additional field)
Digit per line	15 + 4 + 5 + 6 + 6 (4)
Digit dimension (LxH)	50x70 mm/1.97x2.76"
Pixels type	High brightness LEDs
Pixels colour	Yellow (standard) Red (optional)
Step	10 mm/0.39"
Autorange	only outdoor version
Brightness regulation	Automatic by crepuscular sensor

DPY4 - Display Data

Number of lines	4
Line dimension (LxH)	425x30 mm/16.73x1.18"
Pixels type	High brightness LEDs
Pixels colour	Red
Number of pixels	2688
Display information	
Row 1, 2, 3	User programmable
Row 4	Rolling / Scrolling programmable message

Dimensions (mm)



TEMPSOL, IKE20001K, CELLSOL200

Environmental sensors

Carlo Gavazzi, whose experience in the sensors field is well known, has developed these 3 sensors specifically for use in renewable energy applications.

The TEMPSOL has a special design for the measurement of temperatures on flat surfaces, such as the surface of photovoltaic modules.

The IKE20001K sensor is more specifically for open air temperature measurements. It has been designed to be suitable for any kind of environment, resistant to chemical aggression and, being IP67, it can also be used in liquid environments.

Silicon irradiation sensors CELLSOL200 should be commonly used in all monitored PV plants in order to determine precisely the amount of energy the sun is delivering per square metre at any time.

CELLSOL200 is a mono-crystalline silicon cell that provides an output voltage, proportional to the sun irradiation.

All environment sensors are very easy to use and they can be connected directly to Carlo Gavazzi Eos Array modules, in order to provide information to the Eos Array system, or connected to Carlo Gavazzi UDM40 or UDM60 in order to display the values directly.



Detect irradiation and temperature

• Precise output

Both temperature sensors have a tolerance on the reading of 0.3°C or 32.54°F and Class B accuracy according to EN60751/DIN43760.

Each CELLSOL200 sensor has been tested and calibrated individually, by Carlo Gavazzi, with a Sun Simulator Solar Constant 1200 with an ISE calibrated reference sensor. It will provide 75 mV at 1000 W/m of sun irradiation.

The calibration value is printed on the device for the user to be able to match it precisely with the instrument which this sensor will be connected to.

• All weather and long term reliability

All sensors can be used in any weather conditions and the CELLSOL200 with any inclination angle.

The materials used for their construction, the UV resistant cable, and the simple design with no electronic circuitry, guarantee long term durability and reliability.

• Easy Installation

TEMPSOL has a special self adhesive material on one side, which is also a good thermal conductor, and facilitates easy installation and temperature reading reliability.

CELLSOL200 is very simple and easy to install, with only one screw for mechanical installation. There is no need for a power supply, and only two wires for the signal.

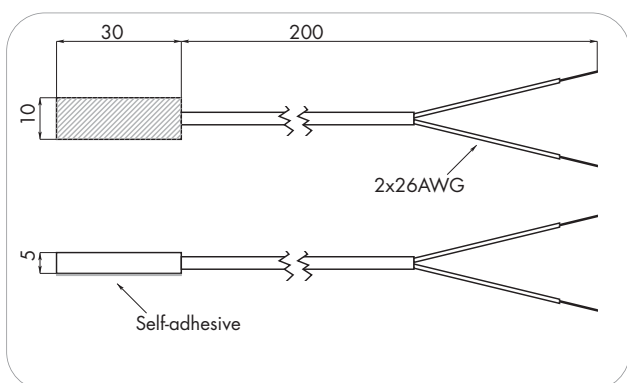
TEMPSOL - General Data

Sensor Pt100 and PT1000	-30°C to +150°C (200°C max.) -22°F to +302°F (392°F max.)
Tolerance	0°C ±0.3°C 32°F ±0.3°F
Accuracy	Class B according to EN60751/DIN43760

TEMPSOL - Mechanical Data

Sensor type	Thermoresistance Pt100 or Pt1000
Mounting	Selfadhesive 30x10 mm
Housing	Encapsulate aluminium
Connection type	2 wires
Connection diameter	26 AWG
Wire length	2 m / 6.56 ft
Insulating material	White twisted teflon

TEMPSOL - Dimensions (mm)



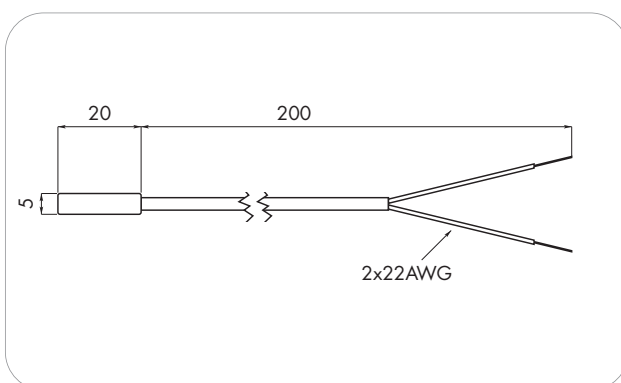
IKE20001K - General Data

Sensor	Pt 1000Ω @ 0°C acc. to IEC 751
Tolerance	0°C ±0.3°C 32°F ±0.3°F
Accuracy	Class B according to EN60751/DIN43760

IKE20001K - Mechanical Data

Sensor type	Thermoresistance Pt1000
Housing	Encapsulate thermoplastic
Connection type	2 wires
Connection diameter	22 AWG
Wire length	2 m / 6.56ft
Insulating material	Copper stranded wire isolated by TPE external diameter ~3.5 mm / 0.14"

IKE20001K - Dimensions (mm)



CELLSOL200 - General Data

Range	from 0 to 1500 W/m ²
Accuracy	±5%, annual average
Output	Aprox. 75 mV @ 1000 W/m ² (true value is reported on the label)

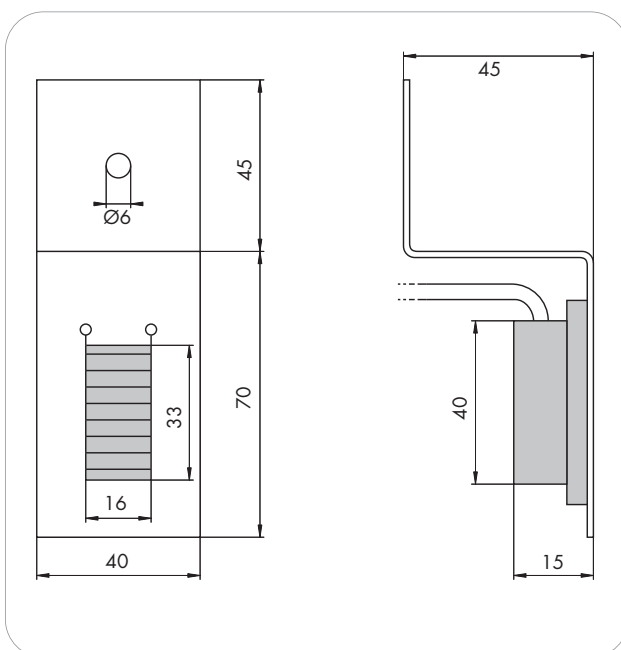
CELLSOL200 - Mechanical Data

Main structure	Anodized aluminium
Cable dimensions	100 x 40 x 40 mm UV proof 3 m
Fixing	1 hole Ø6 mm
Weight	~200 g (with cable)
Warranty	2 years

CELLSOL200 - Sensor Data

Sensor type	Mono-crystalline cell
Mounting	Novaflon and EVA sheet
Cell dimension	16 x 33 mm

CELLSOL200 - Dimensions (mm)



DWS-D & DWS-V

Wind Vanes and Anemometers

Wind speed and direction are extremely important parameters both for a power generating wind turbine and a photovoltaic plant.

Accurate measurements are necessary to keep the power generation and the wind turbine structure under full control. In a PV plant it is important to keep solar trackers safe when wind speed exceeds a fixed limit.

The wind speed sensor DWS-V-DAC provides accurate measurements for a wide variety of applications including wind turbines, buildings, cranes, weather stations, greenhouses etc.

The sensor is designed with a special dust and humidity seal, which makes the sensor impervious to all weather conditions.

The sensor also contains selectable PNP and NPN open collector outputs where a fixed current is switched pro-

portionally to the air speed at a rate of 10 impulses per m/s. A built-in self-regulating heater is standard, reducing the risk of ice; as it is supplied separately, the heating feature can also be controlled separately.

The relative direction sensor DWS-D-DAC is mainly designed for the wind turbine industry.

It registers and signals deflections in the wind direction from a fixed starting point. The sensor is designed with the same features as the wind speed sensor: Dust and humidity sealing, selectable PNP and NPN outputs and a self-regulating heater.



Provide accurate measurements

• Wind Vane

The DWS-D contains both PNP and NPN open collector outputs, in which a fixed current is switched according to the selected direction.

A built-in self-regulated heater reduces the risk of the unit icing-up. The heater is supplied separately, which makes it possible to control the heating.

The DWS-D is equipped with a specially designed protection mechanism, which protects the bearings and the electronic parts against dust and humidity.

The body of the sensor is made of black PVC, and the rotor is manufactured in stainless steel.

- Wind vane for relative wind direction
- Opto-electronic detection
- Indication of 0° and 90° intervals
- Indication of 0° and 90° intervals
- 10 to 28 VDC supply voltage
- All inputs and outputs are protected against reverse polarity and transients

• Cup Anemometer

The DWS-V contains both PNP and NPN open collector outputs, in which a fixed current is switched proportionally to the air speed at the rate of 10 pulses per m/s.

A built-in self-regulated heater reduces the risk of icing-up. The heater is supplied separately, which makes it possible to control the heating.

The DWS-V is equipped with a specially designed protection mechanism, which protects the bearings and the electronic parts against dust and humidity.

The body of the sensor is made of black PVC, and the rotor is manufactured in stainless steel.

- Anemometer with opto-electronic detection
- Measuring range: 2 to 30 m/s
- Current source outputs
- 10 to 28 VDC supply voltage
- All inputs and outputs are protected against reverse polarity and transients
- High ESD protection

DWS-D & DWS-V - General Data

Materials

Body	Black PVC
Rotor	Stainless steel (AISI 303), black painted
Bearings	Instrument ball bearings, stainless steel
Cable	13 m Shielded grey PVC, 8 x 0.25 mm ²

Rotor/housing tightening

Environment	Dust labyrinth
Degree of protection	IP54
Ambient humidity	0 to 100% RH
Climatic protection	Against high humidity, salt and dust

Ambient temperature

Operating temperature	-20 to 60°C (-4 to +140°F)
Storage temperature	-20 to 60°C (-4 to +140°F)

Heating system

Heater	PTC-element
Supply voltage	12 to 24 VAC/DC on separate wires
	Only DWS-D-DDC13 electromagnetic fields thread.

Inrush current	1.5 A
Power consumption	@ -20°C (-4°F): app. 10 W @ +20°C (+68°F): app. 5 W @ +60°C (+140°F): app. 1.5 W

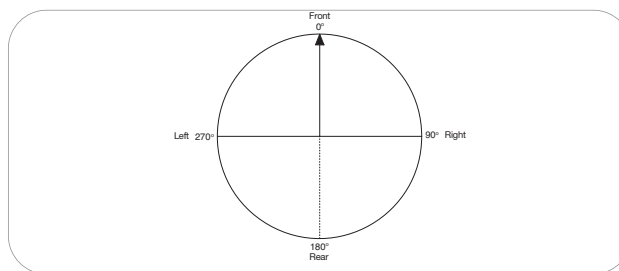
Mounting instruction

	Mounted vertical with M28 thread.
	Only DWS-D marking (dot) on the housing indicates 0° position.

DWS-D - Signal

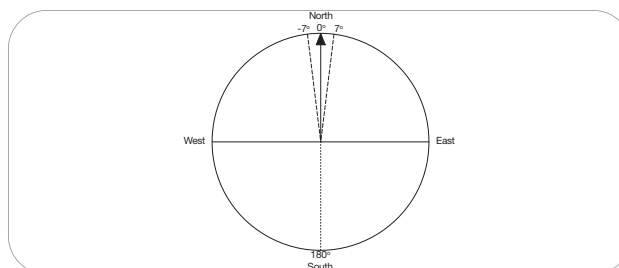
DWS-D-DAC13

wind direction	Bit 1	Bit 2
0° to 90°	0	1
90° to 180°	0	0
180° to 270°	1	0
270° to 360°	1	1
0°	Alternation between 0-1	

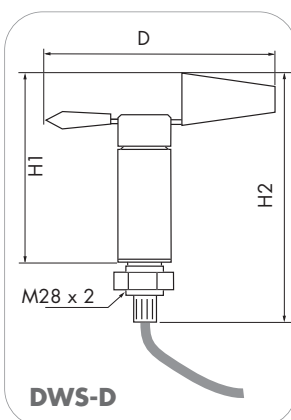


DWS-D-DDC13

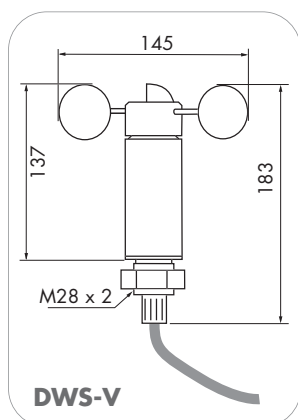
wind direction	Bit 1	Bit 2
0° to 7°	0	1
7° to 180°	1	1
180° to -7°	1	0
-7° to 360°	0	0



Dimensions (mm)



DWS-D



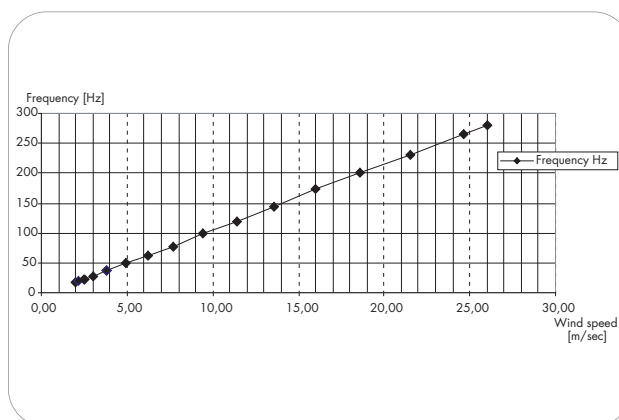
DWS-V

H1, H2 x D mm

DWS-D-DDC13
137, 183 x 145

DWS-D-DAC13
162, 207 x 174

DWS-V - PV Output vs Wind Speed



Notes

Notes



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